

Superconducting Nanowire Single-Photon Detectors for Quantum Information Science

DISSERTATION

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BY

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Abstract

Quantum information science is rapidly advancing and it is likely that society will experience a major shift in the way we handle information in the coming years. As critical advancements such as quantum computation and quantum cryptography progress, the full manifestation of a second quantum revolution is on the horizon. Photons will be an integral part of quantum information science, especially for communication purposes, so we will need to develop excellent single-photon detector technologies.

In this thesis, I describe a promising single-photon detector candidate, superconducting nanowire single-photon detectors, and provide a new theoretical basis for understanding these detectors. Specifically, I explore the relationship between various detector parameters and the readout signal and show that its rising edge can be described using a characteristic time $t_{\text{ch}} \propto \sqrt{\ell/nI_b}$, where ℓ is the length of the detector, n is the absorbed photon number, and I_b is the bias current. Therefore, this work provides a theoretical basis for photon-number resolution in a conventional single-pixel superconducting nanowire single-photon detector. Based on these predictions, I describe a demonstration of photon-number resolution in superconducting nanowire single-photon detectors that correctly identifies more than 99.7% of $n = 1$ events and more than 98% of $n > 1$ events when distinguishing between $n = 1$ versus $n > 1$, as well as explore the effects of other parameters via experiment and by drawing from the literature.

While seeking to understand the quality of the demonstrated photon-number resolution, I find that there is likely some spatial variation in the intrinsic detector parameters. I predict that this may result in some below- T_C resistance in these detectors and verify this experimentally. The resistance ranges from $\sim 50 \Omega$ at $T \sim 0.9 \text{ K}$ to multiple $\text{k}\Omega$ closer to T_C . This is small enough to allow the detectors to function at low temperatures; however, its presence is expected to deteriorate detector performance at higher temperatures via induced latching.

This work constitutes an important addition to the understanding of superconducting nanowire single-photon detectors. The theoretically predicted and experimentally verified photon-number

resolution will increase their versatility and greatly simplify or make possible many quantum information science applications, including linear-optics quantum computing and boson sampling. The measured below- T_C resistance is unexpected and undesirable since it can negatively affect detector performance. This is an area that warrants further exploration.

Dedication

This thesis is dedicated to my mom.

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Contents

- Abstract iii**
- Dedication v**
- Acknowledgements vi**
- Vita viii**
- List of Figures xiii**
- List of Tables xx**
- 1 Introduction 1**
 - 1.1 Quantum Information Science 1
 - 1.2 Qubits 2
 - 1.2.1 Trapped Ions 3
 - 1.2.2 Superconducting Qubits 3
 - 1.2.3 Photons 4
 - 1.2.4 Outlook 5
 - 1.3 Dissertation Roadmap 5
 - 1.4 This Disseration in a Nutshell 6
- 2 Single-Photon Detectors 12**
 - 2.1 Desirable Single-Photon Detector Qualities 12

2.2	Single-Photon Detectors	14
2.2.1	Single-Photon Avalanche Photodiodes (SPADs)	14
2.2.2	Transition Edge Sensors (TESs)	15
2.2.3	Superconducting Nanowire Single-Photon Detectors (SNSPDs)	15
2.3	Introduction to SNSPDs	16
2.3.1	Properties	18
	Critical Temperature	18
	Switching Current	19
	Kinetic Inductance	19
2.3.2	Fabrication	19
2.3.3	Basic Detection Mechanism	20
2.4	Photon-Number Resolution (PNR)	20
2.5	Conclusion	22
3	Superconducting Nanowire Single-Photon Detector Dynamics	24
3.1	Introduction	24
3.2	Qualitative Description of Detection Event	26
3.3	Detection Model	28
3.4	Characteristic Time and Resistance Scales	29
3.4.1	Rescaling Technique	29
3.4.2	Rescaling SNSPD Model	31
3.5	Specifying Phase Front Velocity	33
3.6	Kinetic Inductance Considerations	35
3.6.1	Origin of Kinetic Inductance	35
3.6.2	Effect of Current-Dependent Kinetic Inductance	36
3.7	Experimental Predictions	38
3.8	Conclusion	38

4	Experimental Verification of Electro-thermal Model	40
4.1	Introduction	40
4.2	Demonstration of Photon-Number Resolution in SNSPDs	41
4.2.1	Experimental Setup	41
4.2.2	Results	42
4.3	Effect of Photon Number on SNSPD Readout Pulses	44
4.3.1	Effect of the Finite Bandwidth of the Detection System	46
4.4	Effect of Detector Length on SNSPD Readout Pulses	48
4.5	Application of Universal Model to Experimental Results	49
4.6	Determining Detector Parameters	54
4.6.1	Steady State Current	54
4.6.2	Kinetic Inductance	55
4.6.3	Dark Counts	56
4.7	Conclusion	58
5	Ramifications of Spatial Variations in SNSPD Nanowire Parameters	59
5.1	Introduction	59
5.2	Expected Noise for SNSPD PNR	59
5.2.1	Potential Causes of Increased Noise in SNSPD PNR	62
5.3	Conclusion	63
6	Below-T_C Resistance in SNSPDs	64
6.1	Introduction	64
6.2	SNSPD Resistance Measurements Using a Lock-in Amplifier	64
6.2.1	Experimental Setup and Procedure	65
6.2.2	Initial Results	67
6.2.3	Circuit Model	68
6.2.4	Measuring Capacitance	70

6.2.5	Below- T_C Resistance Results	70
6.3	Theoretical Basis for a Below- T_C Resistance in SNSPDs	73
6.4	Ramifications of Below- T_C Resistance on Detector Functionality	76
6.5	Conclusion	80
7	Conclusions and Future Research	81
	Bibliography	83
A	Cryocooler System	89
A.1	Overview	89
A.1.1	Cryostat and Compressor	90
A.1.2	Vacuum System	91
A.1.3	Sensors/Readings Available	92
A.2	Operating Procedure	93
A.3	Tips and Tricks	94
	Making Good Thermal Contact	95
A.3.1	Faster Warm Up	95
A.3.2	Slower Warming from 850 mK to 4 K	96
A.4	General Maintenance	96

List of Figures

1.1	Simple quantum network example between a trapped ion computer and a superconducting qubit computer. Figure adapted from Monroe and Kim, 2013 and Arute <i>et al.</i> , 2019. Figure from Ch. 2.	6
1.2	(a) Depiction of an SNSPD meander with incident photons. (a-i) Illustration of a region of the SNSPD near an incident and absorbed photon. (a-ii:v) The progression of hot-spot formation, growth, and decay, which is described using the front velocity v . (b) Lumped-element circuit diagram of an SNSPD and readout. (c) An example SNSPD readout pulse. The vertical gray line marks $t = 0$. Figure from Ch. 3.	8
1.3	(a) Rising edges of $n = 1, 2, 3$ traces (gray, blue, red) and (b) with $t_N = t/\sqrt{n}$, as well as their derivatives with respect to time (c) and (d). Figure from Ch. 4	9
1.4	Comparison of expected (grey dashed) vs. measured (blue) SNSPD PNR distribution. Also shown are Gaussians fit to each node of the distribution (orange) and their sum (red). Figure from Ch. 5.	10
1.5	Measured resistance for (a-b) A45 and (c-d) N1 as a function of temperature. Figure from Ch. 6.	11
2.1	Simple quantum network example between a trapped ion computer and a superconducting qubit computer. Figure adapted from Monroe and Kim, 2013 and Arute <i>et al.</i> , 2019.	13

2.2	(a) Scanning electron micrograph (SEM) of optically sensitive region of SNSPD device. (b) Zoomed in SEM of SNSPD nanowire. (c) Illustration of SNSPD absorbing multiple photons.	17
2.3	Photon-detection system based on SNSPDs operating within a cryostat. (a) Under normal operating conditions, the system is under high vacuum and is cooled to ~ 1 K. The SNSPDs are located at the lowest stage of the cryostat, as shown in (b) and (c).	18
2.4	Simple electrical model of SNSPD and readout circuit.	21
2.5	(a) Example of one way PNR could manifest in the readout signal of a detector. (b) Example of how Geiger-like detectors read out a multi-photon signal.	22
3.1	(a) Depiction of an SNSPD meander with incident photons. (a-i) Illustration of a region of the SNSPD near an incident and absorbed photon. (a-ii:v) The progression of hot-spot formation, growth, and decay, which is described using the front velocity v . (b) Lumped-element circuit diagram of an SNSPD and readout. (c) An example SNSPD readout pulse. The vertical gray line marks $t = 0$	27
3.2	Comparison of the exact (Eqs. 3.14 and 3.15) and universal model solutions (Eqs. 3.20 and 3.21, dashed line) for current (a) and resistance (b). Here, the parameters for the exact model are: (blue line) $n = 1$, $I_b = 12.5 \mu\text{A}$; (red line) $n = 3$, $I_b = 12.5 \mu\text{A}$, and (green line) $n = 1$, $I_b = 6 \mu\text{A}$. For all curves, $R_L = 50 \Omega$, $L_k = 0.824 \mu\text{H}$, $v_0 = 60 \text{ pm/ns}$, $R_{sq} = 461 \Omega$, $w = 70 \text{ nm}$, and $I_{ss} = 1.04 \mu\text{A}$	35
3.3	Predicted SNSPD readout pulses for $n = 1$ (grey), 2 (blue), and 3 (red) based on numerically integrating Eqs. 3.1, 3.2, and 3.19. (a) Full pulses including their recovery. (b) Rising edges of the pulses. (c) Derivatives of the rising edges of the pulses. For all curves, $I_b = 12.5 \mu\text{A}$, $R_L = 50 \Omega$, $L_k = 0.824 \mu\text{H}$, $v_0 = 60 \text{ pm/ns}$, $R_{sq} = 461 \Omega$, $w = 70 \text{ nm}$, and $I_{ss} = 1.04 \mu\text{A}$	39
4.1	Schematic of the source setup used in generating multi-photon wavepackets. . .	41

4.2	Schematic of the experimental setup used for collecting the waveform data. The SNSPD is shown as the typical lumped-element depiction with a kinetic inductance L_k and a hot-spot resistance R_N that is a function of the absorbed photon number n	43
4.3	Histogram of differentiated peak heights (blue) for $I_b=11 \mu A$. We fit a sum of Gaussian functions (red) of the form $\sum_{n=1}^3 a_n e^{-((x-b_n)/c_n)^2}$ to the distribution. Resulting individual Gaussians (orange curves) and their centers (gray vertical lines) are also shown. Here, $a_1 = 0.95 \pm 0.01$, $b_1 = 57.38 \pm 0.04$, $c_1 = 4.59 \pm 0.06$, $a_2 = 0.355 \pm 0.009$, $b_2 = 77.7 \pm 0.4$, $c_2 = 8.1 \pm 0.4$, $a_3 = 0.095 \pm 0.009$, $b_3 = 93 \pm 1$, and $c_3 = 8 \pm 2$	44
4.4	Histogram of differentiated peak heights (blue) for $I_b=11 \mu A$. We fit a sum of Gaussian functions (red) of the form $\sum_{n=1}^3 a_n e^{-((x-b_n)/c_n)^2}$ to the distribution with the constraint that the area under each individual Gaussian (orange curves) obeys Poisson statistics. Here, $a_1 = 0.95 \pm 0.01$, $b_1 = 57.38 \pm 0.04$, $c_1 = 4.54 \pm 0.06$, $a_2 = 0.34 \pm 0.04$, $b_2 = 77.2 \pm 0.6$, $c_2 = 7.9 \pm 0.5$, $a_3 = 0.10 \pm 0.02$, $b_3 = 91 \pm 3$, and $c_3 = 11 \pm 3$	45
4.5	(a) Rising edges of $n = 1, 2, 3$ traces (gray, blue, red) and (b) with $t_N = t/\sqrt{n}$, as well as their derivatives with respect to time (c) and (d).	47
4.6	Prediction that $t_{\text{rise}} \propto \sqrt{\ell}$ (red curve) compared with theoretical predictions from Smirnov <i>et al.</i> Smirnov <i>et al.</i> , 2016 (blue dots), alongside their experimental results for NbN on Si/Si ₃ N ₄ (green diamonds) and Si/SiO ₂ (orange squares). We fit their theoretical predictions for values $> 20 \mu m$ (gray vertical line).	49
4.7	Experimental values of t_{ch} (blue dots) extracted by fitting with the linear model vs. (a) n with $I_b=11 \mu A$ and (b) I_b with $n = 1$, as well as their fits (red curves) to the model $t_{\text{ch}} = A/n^a$ or $t_{\text{ch}} = B/(I_b)^b$ respectively.	50
4.8	Universal model fits (red solid line) for $n = 1, 2$, and 3 (blue points).	52
4.9	Universal model fits (red solid line) for different I_b (blue points).	53

4.10	Measurement schematic (a) and resulting IV curve (b) for the Quantum Opus detector. There is a $1\text{ M}\Omega$ series resistor R_{bias} between the voltage source and the detector, resulting in a slope of $1\text{ M}\Omega$ during the superconducting portion of the IV curve. The normal resistance is about $6.53\text{ M}\Omega$ including R_{bias} . The bias-T is a Mini-Circuits ZFBT-6GW and the low pass filter (LPF) is a Mini-Circuits SLP-1.9+.	54
4.11	Measurement schematic (a) and resulting SNSPD pulse (b). We fit Eq. 4.2 (red) to the falling edge (light blue region) and extract L_k . The low pass filter (LPF) is a Mini-Circuits SLP-1.9+.	55
4.12	The detector dark count rate (points with solid line) varies from a few counts per second (cps) for low I_b , up to $> 8,000$ cps for high I_b . The total count rate (squares with dotted line) is also shown. (Note measured results are given by plot points – the lines are just guides for the eye.)	56
4.13	Comparison of multi-photon absorption data (blue) with dark count rates (red) at $I_b = 12.8\mu\text{A}$. Both distributions are normalized such that the area under their curves are unity, then the dark counts are further scaled by the ratio of the DCR over the overall count rate during data collection. Gaussians are fit to the $n = 1$ peak and their centers are shown as vertical lines for both source (solid blue) and dark (dotted red) counts.	57
5.1	Comparison of expected (grey dashed) vs. measured (blue) SNSPD PNR distribution. Also shown are Gaussians fit to each node of the distribution (orange) and their sum (red) as previously shown in Fig. 4.3	61
6.1	Measurement setup for measuring SNSPD resistance using a lock-in amplifier. Not picture is the lock-in load resistor which is specified at $100\text{ M}\Omega$ and therefore has a negligible effect on V .	66

6.2	(a) Voltage and (b) phase measured by the lock-in amplifier using 10, 50, and 100 kHz driving frequency for detectors A45 (blue, red, green respectively) and N1 (purple, pink, orange respectively).	68
6.3	Voltage and phase measured by the lock-in amplifier at 10, 50, and 100 kHz driving frequency for detectors (a-b) A45 (blue, red, green respectively) and (c-d) N1 (purple, pink, orange respectively) at low temperatures (below 2.5 K).	69
6.4	Measured $ V_C $ as a function of frequency (blue dots) fit with Eq. 6.4 (red line). . .	70
6.5	Interpolation (red curve) of measured $ V(2\pi f, T) $ (blue dots) for $f = 50$ kHz for A45 as a function of temperature.	71
6.6	Measured resistance for (a-b) A45 and (c-d) N1 as a function of temperature. . . .	72
6.7	Predicted phase based on measured resistance (curves) vs. measured phase (points) at 10, 50, and 100 kHz for (a) A45 (blue, red, green respectively) and (b) N1 (purple, pink, orange respectively) as a function of temperature.	73
6.8	Examples of stable hot-spot regions (a) in a long, thin nanowire and (b) at the interface between a microbridge and a bulk superconductor. Figure based on Gurevich <i>et al.</i> (Gurevich and Mints, 1987).	75
6.9	Critical temperature of MoSi thin films as a function of Si composition. 50 nm (square points) and 200 nm (diamond points) data taken from Kubo (Kubo, 1988). 100 nm (circular points) taken from Bosworth <i>et al.</i> (Bosworth <i>et al.</i> , 2015).	76
6.10	Lumped-element circuit model of SNSPD with anomalous below- T_C resistance R_a and AC coupled readout.	77
6.11	(a) Hot-spot resistance and (b) readout current for a detection event in an AC-coupled SNSPD with $R_a = 0$ (grey dashed), 100 (red), and 200 (blue) Ω . Here, $R_L = 50 \Omega$, the AC coupling capacitor C is 400 pF, $I_b = 10 \mu A$, and the SNSPD has $L_k = 1 \mu H$, $v_0 = 50$ m/s, $I_{ss} = 1.04 \mu A$, normal sheet resistance $R_{sheet} = 461 \Omega$, and width $w = 70$ nm.	78

6.12 A45 readout pulses for various operating temperatures with $I_b = 0.9I_{\text{sw}}$. The detector stops reading out pulses above $T = 2.24$ K.	79
A.1 Components of the cryocooler system: (a) cryostat, (b) vacuum pump, (c) compressor.	90
A.2 Vacuum valves in the cryostat system. (a) A valve is present at the top of the cryostat that connects it to the rest of the vacuum system. (b) Two venting valves are present.	92

List of Tables

4.1	Values of t_{ch} found by fitting experimental waveforms to the universal model solution.	51
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Chapter 1

Introduction

1.1 Quantum Information Science

Quantum information science (QIS) is a rapidly expanding field that promises to bring about significant advances in the ways we extract, handle, and transmit information. By exploiting the principles of quantum mechanics, QIS will deliver better quality information faster and more securely. This will have wide reaching affects in many areas of society, including within government, military, and private sector affairs throughout the world.

Pioneered by physicists, the exploration of quantum mechanics revolutionized the 20th century by demystifying the physical laws of matter on the microscopic scale. These advancements in our understanding of the world brought about many inventions that are critical to modern life, including the transistor and the laser. As a result, this sequence of events is often referred to as the first quantum revolution. In the 1970s and 1980s, researchers became interested in applying the concepts of quantum mechanics to information theory. Since then, the field of QIS has exploded in popularity and promising advancements, and we are widely considered to be squarely in the middle of the second quantum revolution. Although, QIS is an interdisciplinary field, combining the expertise of physicists, engineers, computer scientists, and mathematicians; physicists once again are playing an integral role in understanding the fundamental concepts underlying QIS and demonstrating important experimental milestones in the

field.

The most widely known subspace of QIS is quantum computing. Due to the fact that a quantum computer will significantly outperform a classical computer at certain tasks including factoring of prime numbers, quantum computing has the potential to render widely used encryption schemes (notably Rivest-Shamir-Adleman encryption) based on prime number factorization useless (Shor, 1999). Further, and perhaps more short term, quantum computing will allow for the simulation of materials at a quantum level. This important advancement will have wide reaching impacts on many facets of society, including allowing for improved drug design, advances in sustainable power generation, development of new materials, as well as advancements in fundamental physics (Cao, Romero, and Aspuru-Guzik, 2018; Bauer *et al.*, 2020; Bassman *et al.*, 2020; Ajagekar and You, 2019; Alexeev *et al.*, 2021).

Another important subspace of QIS is quantum cryptography. This field works towards securing information in quantum mechanical ways, which promises vastly improved security over our current methods of data encryption (Bennett, 1992). As quantum computing rapidly advances, quantum cryptography is especially important because it will provide alternate ways of securing information that would be robust against quantum computing based security attacks. Between the rise of quantum computing and the complementary advent of quantum cryptography, QIS promises to completely reshape our information security landscape in the coming years.

1.2 Qubits

The main advantage that is gained by QIS over classical information theory is the use of quantum bits (qubits). Qubits are quantum mechanical two-state systems that function similarly to classical bits in that they can represent both a zero or a one for information storage and manipulation. However, because qubits are quantum mechanical in nature, they can be placed in

superposition states which functionally drastically increases their computational efficiency for many applications (Barnett, 2009).

Although any two-state quantum system is by definition a qubit, there are a number of specific systems that are emerging as promising candidates for use in QIS. Here I briefly describe the basic principles and current state of the art for trapped ions, superconducting qubits, and photons. The quality of a qubit is typically measured by decoherence time (how long can a qubit state be stored before it is lost) and scalability, which I discuss for each of these candidates below.

1.2.1 Trapped Ions

Trapped ion based qubits are constructed by confining and suspending individual ions using electromagnetic fields. Once trapped, the valence electron energy level of an ion can be manipulated using laser light to form the desired qubit state (Wineland *et al.*, 1998). Trapped ions make very reliable qubits, with decoherence times on the order of seconds, but they are typically considered less scalable, with the current state-of-the-art trapped-ion based quantum computers having tens of qubits (Wright *et al.*, 2019).

1.2.2 Superconducting Qubits

Superconducting qubits are superconducting circuits which are carefully designed to behave as quantum anharmonic oscillators through the use of Josephson junctions (Krantz *et al.*, 2019). Due to their anharmonicity, it is possible to create two state systems which are able to function as qubits. Superconducting qubits typically have decoherence times of microseconds, but groups have made large strides in terms of scalability, with demonstrations of up to 65 qubits being reported (Cho, 2020; Burnett *et al.*, 2019; Arute *et al.*, 2019).

1.2.3 Photons

Photons make promising qubits for many applications for a wide variety of reasons. There are many degrees of freedom inherent to a photon that allow for multiple ways to encode a qubit, including through polarization, time-frequency binning, photon-number, and angular momentum (Northup and Blatt, 2014). Additionally, photons make extremely reliable qubits that survive over long distances, which corresponds to long times via the speed of the photonic wavepacket. In vacuum, the coherence distance is set practically by diffraction, which causes the transverse extent of the photonic wavepacket to spread with propagation distance. There is a practical limit of the aperture of the optical receiver, which sets the fraction of received photonic wavepackates.

Optical waveguides, such as optical fibers, prevent diffractive spreading, but decoherence arises from other mechanisms due to the coupling of light to matter. One form of decoherence is loss of photons. Photons are lost from waveguides via Rayleigh scattering (dominant for shorter wavelengths) and molecular absorption (at longer wavelengths). The lowest-loss band in telecommunication-grade fibers is 1550 nm, which is the cross-over from where Rayleigh scattering and molecular absorption contribute equally to the loss (Essiambre *et al.*, 2010). For a modern optical fiber, the loss is 0.1 dB/km at 1550 nm, allowing for practical quantum communication of 10's to 100's of km. In this thesis, I focus on this wavelength band because of the lower loss and because of the availability of classical optical telecommunication components for manipulating the quantum state of photons.

Other source of decoherence arises from spreading of photonic wavepackets due to dispersion in the index of refraction of the glass. This is a deterministic process and can be reversed using dispersion-compensation modules. Time-dependent polarization changes also occur in a fiber due to temperature changes or flexing or bending of the fiber, such as that due to traffic moving near a buried optical fiber. These changes tend to be on the few millisecond time scale and can be corrected using feedback methods.

1.2.4 Outlook

In the future, we will likely have many different types of quantum systems using different qubit architectures due to the differing strengths and weaknesses of the various technologies. This will make interfacing between devices, in other words allowing for some kind of quantum internet, essential. Unlike the other qubits discussed thus far, photons are able to transport quantum states over large distances, which is extremely important for any sort of connectivity between future quantum devices. Further, photons travel at the speed of light, which allows for the fastest transmission of information possible. Additionally, unlike competing qubit technologies, optical photons do not require cryogenic temperatures, and therefore can be a useful bridge to connect cryogenic quantum systems. Finally, light is already used for classical communication as well as in many optics laboratories throughout the world. In using photons for quantum communication, we can make use of that existing infrastructure and make a much easier transition into the quantum realm.

1.3 Dissertation Roadmap

Thus, for the future of quantum communication or quantum computation, it is extremely important that we develop good tools for working with photons. In this thesis, I focus specifically on detecting photons. In the remainder of this Introduction, I provide a brief overview of the key points I plan to communicate throughout this thesis. In Ch. 2, I discuss desirable qualities of a single-photon detector, as well as explore the current state of the art for photon detection and introduce the detectors I focus on in this thesis, superconducting nanowire single-photon detectors (SNSPDs). In Ch. 3, I present a theoretical framework for understanding photon detection in SNSPDs, which includes predictions for how the output of an SNSPD changes based on various factors, including absorbed photon number. In Ch. 4, I discuss experimental verification for the predictions made in the preceding chapter, including the first experimental demonstration of photon-number resolution (PNR) in SNSPDs. In Ch. 5, I take a closer

FIGURE 1.1: Simple quantum network example between a trapped ion computer and a superconducting qubit computer. Figure adapted from Monroe and Kim, 2013 and Arute *et al.*, 2019. Figure from Ch. 2.

look at the quality of PNR demonstrated, and this prompts a discussion on the effect of spatial variations in detector parameters. In Ch. 6, I present experiments conducted to try to understand anomalous below- T_C resistance that was detected throughout the course of this work that may be related to spatial variations in detector parameters, as well as lay down the theoretical foundation to understand below- T_C resistance in SNSPDs. Finally, in Ch. 7, I explore future scientific pathways and potential experiments based on this work. Much of the work presented in this thesis was done collaboratively. At the start of each section I include a brief description of who was responsible for each component, highlighting my contributions.

1.4 This Dissertation in a Nutshell

In this section, I give a high level overview of my thesis. By highlighting one figure from each chapter, I hope to provide a useful summary of the most important points and findings.

The burgeoning field of QIS promises to drastically improve information quality and security throughout many areas of society. Because they are ideally suited for quantum-based information transfer, photons are highly likely to be an integral part of the impending QIS revolution. Fig. 1.1 shows an example of communication between two quantum computers using photons, which will likely form the basis of a quantum internet.

To work with photons, it is extremely important that we develop excellent technologies for

detecting photons accurately, quickly, efficiently, and while exploiting existing infrastructure as much as possible. Superconducting nanowire single-photon detectors (SNSPDs) are an excellent choice because they fulfill all of these criteria handily. Because of this, much work has already gone into developing and understanding SNSPDs.

Here, I expand the body of knowledge on SNSPD dynamics to include PNR, specifically showing that this and other information is encoded in the rising edge dynamics of the SNSPD readout pulse. The origin of this effect can be understood using an electro-thermal model, which combines the electrical dynamics of the SNSPD and readout circuit with the thermal dynamics associated with photon absorption, as shown in Fig. 1.2.

By extracting a characteristic time-scale for the rising edge of the SNSPD readout pulse, I am able to predict that $t_{\text{ch}} \propto 1/\sqrt{n}$, meaning SNSPDs are inherently capable of PNR. This is clearly shown in Fig. 1.3, where increasing photon number decreases the rise time of the readout pulse and increases the derivative of the readout pulse. By rescaling the rising edge of SNSPD pulses using $t_N = 1/\sqrt{n}$, these differences are removed, highlighting this scaling law. Based on this analysis, I develop a universal model for SNSPD rising edge dynamics which can be used to extract information, such as photon number, from the rising edge of SNSPD pulses.

Careful analysis of the distribution of SNSPD readout rise times (maxima of the derivative) further illuminates the inherent PNR capabilities of SNSPDs, as shown in Fig. 1.4. Although this is a very exciting result, we expect better PNR than was demonstrated in the experiment. As one possible explanation for this, I explore the possibility of imperfect superconductivity in the SNSPDs below their critical temperature. I show that this could result in a resistance that persists even when the device is nominally in the superconductive state. Using a lock-in amplifier, I attempt to measure this anomalous resistance in two different SNSPDs. I find that there is a nonzero resistance below the critical temperature in both devices, and that it changes as a function of temperature, as shown in Fig. 1.5. This resistance has not previously been demonstrated to exist in SNSPDs, and warrants further investigation as it has the potential to negatively affect SNSPD operation.

FIGURE 1.2: (a) Depiction of an SNSPD meander with incident photons. (a-i) Illustration of a region of the SNSPD near an incident and absorbed photon. (a-ii:v) The progression of hot-spot formation, growth, and decay, which is described using the front velocity v . (b) Lumped-element circuit diagram of an SNSPD and readout. (c) An example SNSPD readout pulse. The vertical gray line marks $t = 0$. Figure from Ch. 3.

The findings presented in this thesis improve our understanding of SNSPDs and provide exciting new avenues for their use, especially with regards to applications requiring PNR.

FIGURE 1.3: (a) Rising edges of $n = 1, 2, 3$ traces (gray, blue, red) and (b) with $t_N = t/\sqrt{n}$, as well as their derivatives with respect to time (c) and (d). Figure from Ch. 4

FIGURE 1.4: Comparison of expected (grey dashed) vs. measured (blue) SNSPD PNR distribution. Also shown are Gaussians fit to each node of the distribution (orange) and their sum (red). Figure from Ch. 5.

FIGURE 1.5: Measured resistance for (a-b) A45 and (c-d) N1 as a function of temperature. Figure from Ch. 6.

Chapter 2

Single-Photon Detectors

The quality of any quantum communication system is directly affected by the performance of the single-photon detectors used in the system. Throughout this section I discuss desirable qualities in a single-photon detector, including fast and precise timing, high efficiency, low false counts, operation at a useful wavelength, and photon-number resolution (PNR). Then, I briefly discuss some of the popular options for single-photon detection including single-photon avalanche photodiodes, transition edge sensors, and superconducting nanowire single-photon detectors (SNSPDs). Finally, I provide a simple introduction to SNSPDs, including the current strengths and weaknesses of these detectors as of the writing of this thesis.

2.1 Desirable Single-Photon Detector Qualities

Fig. 2.1 shows a simple example of communication between two quantum computers using photons. In this scenario we are interfacing between a trapped-ion based system and a superconducting qubit system; however, any quantum network will likely require photons for long-distance information transfer.

One of the most important metrics in this situation is the speed at which we can transmit information, which is directly proportional to the speed at which a single-photon detector can detect photons. Typically immediately after a single-photon detector detects a photon, there is

FIGURE 2.1: Simple quantum network example between a trapped ion computer and a superconducting qubit computer. Figure adapted from Monroe and Kim, 2013 and Arute *et al.*, 2019.

an associated recovery time during which the detector is not sensitive to new incoming photons. The recovery time puts an upper limit on the rate of information transfer, so it is ideal for it to be as small as possible.

Single-photon detectors also typically have an uncertainty associated with the exact time of arrival of a given photon. There is some variable amount of time between when a photon arrives at the detector and when the detector actually registers the event. This uncertainty is called timing jitter. For many quantum communication protocols, it is important to measure precisely when a photon arrives, so lower jitter is strongly preferred (Islam *et al.*, 2019).

Across all single-photon detectors, there is some proportion of photons which do not trigger a detection event. If a detector misses a photon, it translates directly to a decrease in the information transfer rate. Further, for some quantum cryptography protocols, missing photons could represent a significant security loophole (Lo, Curty, and Tamaki, 2014). Thus, it is desirable for single-photon detectors to have as high efficiency as possible.

Additionally, all single-photon detectors occasionally register detection events when there were no photons present. The origin of these false events varies based on the individual detector technology and the environment in which the detector is operated, but they are all considered to be "dark counts". A high dark count rate can obscure information, as well as waste detector resources by causing the detector to enter a recovery period despite no useful events being detected, ultimately having the effect of slowing down the information transfer rate.

Therefore, lower dark count rates are preferred.

As discussed in Ch. 1, since light is used in classical communication, there is much existing infrastructure that can be used for quantum communication with photons. Most of this infrastructure has been carefully designed to minimize signal loss within a particular range of wavelengths called the telecom bands, the most common of which is centered at 1550 nm. Thus, if we desire to make use of pre-existing infrastructure and minimize signal loss, it is important to make detectors that can detect photons at 1550 nm.

Finally, for many quantum applications it is useful for a detector to be able to count photon number. Practically this means that if multiple photons arrive at the detector simultaneously, it is able to resolve the number of photons and not just the fact that there were photons present. I discuss this in more detail below.

2.2 Single-Photon Detectors

There is a wide array of single-photon detection technologies available (Eisaman *et al.*, 2011). In this section, I briefly discuss three of the most prominent single-photon detectors used in quantum information applications: single-photon avalanche photodiodes, transition edge sensors, and superconducting nanowire single-photon detectors. All of these detectors have strengths and weaknesses, based in part on the desirable detector qualities described above, which make them appropriate for different situations.

2.2.1 Single-Photon Avalanche Photodiodes (SPADs)

SPADs are solid state single-photon detectors that operate similarly to a conventional photodiode that is reverse biased with a very high voltage. When a photon is absorbed, it knocks loose an electron which is rapidly accelerated by the reverse bias to a kinetic energy that exceeds the ionization energy for the SPAD material. As a result, more electrons are ionized and an avalanche process occurs which constitutes a detection event. SPADs are relatively fast, with

maximum count rates in the MHz and timing jitter in the hundreds of picoseconds. Unfortunately, SPADs have low efficiency at about 10% at 1550 nm. Despite this, SPADs are convenient for many applications because they are operated at room temperature. SPADs are not currently understood to have PNR capabilities.

2.2.2 Transition Edge Sensors (TESs)

TESs are a kind of bolometer, detecting incoming photons by measuring small changes in temperature caused by photon absorption. Because they measure extremely subtle temperature fluctuations, TESs must be operated at cryogenic temperatures. TESs boast excellent efficiency of above 90% at 1550 nm, but they are typically very slow, with recovery times on the order of microseconds. Unlike the other detectors discussed in this thesis, TESs are capable of resolving large numbers of photons, with up to tens of photons demonstrated (Lita, Miller, and Nam, 2008).

2.2.3 Superconducting Nanowire Single-Photon Detectors (SNSPDs)

SNSPDs are widely used detectors which use nanowires of superconducting material to detect absorbed photons. When a photon is absorbed by an SNSPD, it disturbs the superconductivity and causes a region of the wire to become normal, and therefore, resistive. This sudden change in resistance can be sensed with an appropriate readout circuit, which I discuss in more detail below.

SNSPDs boast impressive timing characteristics, with recovery times demonstrated at < 1 ns and timing jitter at < 3 ps (Cherednichenko *et al.*, 2021; Korzh *et al.*, 2020). Equally as impressive, multiple groups have developed SNSPDs with efficiencies of up to 99% in the telecom bands (Reddy *et al.*, 2020; Chang *et al.*, 2021). SNSPDs typically have low dark counts and high polarization-insensitive efficiency has been demonstrated (Meng *et al.*, 2020). For all of these reasons, SNSPDs make excellent single-photon detectors for quantum communication.

Unfortunately, due to their superconducting nature, they must be operated in a cryogenic environment, which limits their use in some applications. However, for experiments and protocols which do not require mobility or a small form factor, SNSPDs can be considered one of the best choices available, especially if communication speed is an important factor. SNSPDs have already been integrated with many of the qubit technologies described in Ch. 1, and they have been integral to many of the state-of-the-art quantum communication demonstrations to date (Todaro *et al.*, 2021; Crain *et al.*, 2019; Islam *et al.*, 2019). For these reasons, I focus on SNSPDs throughout this thesis.

2.3 Introduction to SNSPDs

SNSPDs are microscopic devices, typically on the order of $\sim 15 \mu\text{m}$, as shown in Fig. 2.2(a). A single device consists of a very long ($\sim 1 \text{ mm}$), narrow ($\sim 100 \text{ nm}$), and thin ($\sim 10 \text{ nm}$) nanowire that is fabricated in a meander pattern, as shown in Fig. 2.2(b). The dimensions and geometry of the nanowire meander are chosen to optimize the probability that a photon will be absorbed and have sufficient energy to locally disturb the superconductivity of the device. The diameter of the meander is designed to correspond with the output mode of a single-mode optical fiber, which typically have cores of $\sim 5 \mu\text{m}$. The spacing between sections of the meander is typically $\sim 100 \text{ nm}$, such that the entire meander is an effective optical medium for photons in the visible and infrared spectrum. Fig. 2.2(c) shows an illustration of an SNSPD absorbing multiple photons. SNSPD devices are placed in specially designed packaging to allow for the delivery of photons to the detector through an optical fiber and for reading out electrical signals created during a photon absorption event. The packaged SNSPD is placed in a cryogenic system, as shown in Fig. 2.3.

FIGURE 2.2: (a) Scanning electron micrograph (SEM) of optically sensitive region of SNSPD device. (b) Zoomed in SEM of SNSPD nanowire. (c) Illustration of SNSPD absorbing multiple photons.

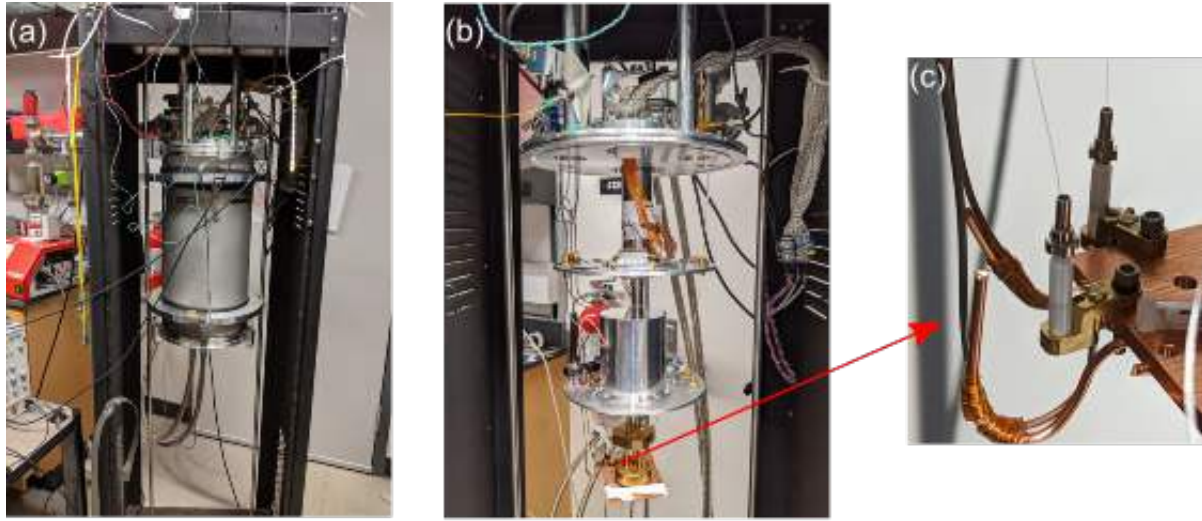


FIGURE 2.3: Photon-detection system based on SNSPDs operating within a cryostat. (a) Under normal operating conditions, the system is under high vacuum and is cooled to ~ 1 K. The SNSPDs are located at the lowest stage of the cryostat, as shown in (b) and (c).

2.3.1 Properties

All SNSPDs have some basic properties which are a result of their superconducting nature (Tinkham, 2004).

Critical Temperature

All SNSPDs have a critical temperature, below which the device is superconducting and above which the device is normal. SNSPDs must be operated below this critical temperature to function. The exact temperature threshold for optimal SNSPD operation depends on the superconducting material used. A wide array of materials are currently being used to make SNSPDs, including NbN, MoSi, and WSi (Dauler *et al.*, 2014). All of these materials have critical temperatures on the order of or less than ~ 10 K, so a He cryostat system is typically required for their operation. An example of such a system is shown in Fig. 2.3. More details about the cryogenic system used throughout this work can be found in Appendix A.

Switching Current

Additionally, all SNSPDs have a limit to the amount of current that can be driven through them before they will switch into the normal state. The origin of this switching current is based in the fact that superconductors have a critical magnetic field, beyond which their superconductivity is destroyed. Furthermore, the switching current of the device is coupled with the operating temperature of the device. As temperature increases, the switching current decreases and it becomes easier for the device to transition into the normal phase. The SNSPDs used throughout this thesis have a switching current on the order of $\sim 10 \mu\text{A}$ at $\sim 1 \text{ K}$.

Kinetic Inductance

When superconductors experience an AC-electromagnetic field, they exhibit inductive behavior. The origin of this inductance comes from the fact that electrons in a superconductor form Cooper pairs and travel without scattering off each other. This results in a significant increase in their kinetic energy when compared with electrons moving in a normal metal. As a result, if there is a sudden change in the field experienced by the superconductor, there is a delay in the response of the electron pairs. This manifests as an inductance, which is called the kinetic inductance. Mathematically, this inductance behaves identically to an electromagnetic inductance. The kinetic inductance is temperature dependent, increasing up to three times its base temperature value when warmed close to its critical temperature (Santavica *et al.*, 2016; Nicolich *et al.*, 2017).

2.3.2 Fabrication

SNSPDs are carefully designed to optimize efficiency at the desired operating wavelength. The dimensions of the nanowire are chosen to allow for a single photon to be enough to cause a localized section of the nanowire to transition to the normal state. To increase the probability that a photon will actually hit the device, the nanowire is patterned in a meander which is sized

to match the output of an optical fiber. The features of an SNSPD are on a nanometer scale, so devices are typically made with electron beam lithography (Natarajan, Tanner, and Hadfield, 2012; Yang, 2009; DeBry, 2020).

2.3.3 Basic Detection Mechanism

SNSPDs can be understood electrically using lumped-circuit elements, as shown in Fig. 2.4. The lumped-element depiction of an SNSPD includes an inductor that represents the kinetic inductance, and a time-dependent resistor that represents the normal state resistance, in series. To capture the superconducting vs. normal switching behavior of the device, the resistor is placed in parallel with a switch. The device is considered superconducting when the switch is closed, and normal when the switch is open.

The detector is placed in parallel with a readout circuit, which is typically an amplifying circuit with a small load impedance of $50\ \Omega$. In the absence of a photon, the switch is closed and any bias current travels through the device to ground. As a result, nothing is seen at the readout. However, when a photon is absorbed, the switch is open and there is a large normal resistance in the detector, typically on the order of a few $k\Omega$. This causes current to leave the detector and instead travel through the readout. The signal in the readout arm constitutes a detection event. This process will be described in much greater detail in Ch. 3.

2.4 Photon-Number Resolution (PNR)

A final desirable quality for photon detectors is PNR, which is the ability to know exactly how many photons arrive at the detector simultaneously. Practically, this means that when multiple photons are absorbed by the detector, there is a signal produced that is measurably different from a single-photon signal. An example of PNR which manifests in the amplitude of the detector signal is shown in Fig. 2.5(a). Most single-photon detectors operate in a Geiger counter-like

FIGURE 2.4: Simple electrical model of SNSPD and readout circuit.

state, in which multiple photons incident on the device simultaneously will produce a signal that is identical to a single-photon signal, as shown in Fig. 2.5(b).

PNR is very useful and at times required for a wide range of quantum communication applications. For example, PNR improves security in quantum cryptography protocols, allows for a more robust and widespread quantum communication network based on quantum repeaters, and makes quantum computing protocols based on linear optics quantum computing and boson sampling possible (Jönsson and Björk, 2019).

There have been a number of ways proposed to do PNR. The most straightforward is likely to use TESs, as these are well documented to have excellent PNR of up to 10s of photons. Unfortunately, TESs are slow, which makes them poorly suited to quantum communication applications. An alternative approach is to simply use many single-photon detectors and divide the light signal equally among them. This approach has been used to demonstrate PNR SNSPD arrays (Divochiy *et al.*, 2008). Unfortunately, this approach still has the possibility of information lost when multiple photons end up routed to the same detector, which limits the rate at which it can be used. For example, groups have demonstrated boson sampling with 14 photons, but

FIGURE 2.5: (a) Example of one way PNR could manifest in the readout signal of a detector. (b) Example of how Geiger-like detectors read out a multi-photon signal.

they required 60 single-photon detectors to achieve this (Wang *et al.*, 2019). Further, in the case of SNSPDs, there are significant space and thermal load considerations to consider, which limit the number of devices that can be run simultaneously, especially if cryogenic amplification is used. In Chs. 3 and 4, I will explore the possibility of using only a single-pixel, conventional SNSPD device to achieve PNR.

2.5 Conclusion

Because quality and rate of quantum communication is directly related to our detection capabilities, it is important that we develop excellent single-photon detectors to meet this need. Three promising candidates are SPADs, TESs, and SNSPDs. For many applications, SNSPDs are the best choice due to their excellent speed, precision, and efficiency at useful wavelengths.

One area where they are currently lacking is the ability to do PNR. In the remainder of this thesis, I describe an avenue for PNR with conventional SNSPDs, as well as explore other detector characteristics that can affect the overall quality of photon detection.

Chapter 3

Superconducting Nanowire Single-Photon Detector Dynamics

3.1 Introduction

In this section, I describe an electro-thermal model for understanding superconducting nanowire single-photon detector (SNSPD) dynamics that includes the potential for multiple photons to be absorbed simultaneously by the detector. Based on this model, I derive a characteristic time scale for the rising edge of the SNSPD readout pulse, which contains a dependence on photon number and other detector parameters. This constitutes a prediction that SNSPDs have inherent PNR capabilities. Then, I derive a simpler, universal model that is valid for examining the rising edge of SNSPD readout pulses. The work presented in this section was done by me under the guidance of Daniel Gauthier and Gregory Lafyatis.

A qualitative picture of SNSPD operation involves one or more photons absorbed by the device simultaneously that create resistive regions in the nanowire, known as hot-spots, which divert current out of the detector and into the readout circuit, constituting a detection event. A complete, quantitative model requires knowledge of the spatial-temporal dynamics of the non-equilibrium distribution of quasi-particles in the superconductor and its interaction with the readout electronics. The understanding of the microscopic details of these devices is rapidly

advancing (Engel *et al.*, 2015a); however, theoretical models of SNSPDs frequently have more parameters than measurable constraints (Yang *et al.*, 2007; Kerman *et al.*, 2009). Hence, similar behaviors may be fit using different sets of model parameters, thus potentially obscuring the physical principles underlying SNSPD behavior.

In this section, I describe a simple SNSPD model that captures the essential physics of the link between hot-spot growth and features of the rising edge of the electrical readout pulse. This approach is motivated by a model proposed by Kerman *et al.* and based on past work on the speed of a superconducting-normal wavefront in a long, narrow superconducting wire (Gurevich and Mints, 1987). Previously, the rising edge of the readout signal has been described as an exponential growth with time constant L_k/R_{hs} (Natarajan, Tanner, and Hadfield, 2012), where hot-spot resistance $R_{\text{hs}} \sim 10^3 \Omega$ is constant during the presence of the hot-spot. However, as shown below, this is not an accurate picture and leads to incorrect conclusions about the detector turn-on dynamics.

The electro-thermal model presented here identifies a ‘universal curve’ for the electrical signal, which on its own constitutes a valid universal model for the rising edge of SNSPD readout pulses. Converting the universal model to physical units requires using only two scale parameters, which can be determined experimentally, and gives a simple relation between the microscopic SNSPD parameters and the readout signal. Through its inherent scaling relations, the model explains the recent demonstration of multi-photon resolution in a conventional SNSPD (Cahall *et al.*, 2017). Further, it predicts the scaling of the turn-on time with nanowire length ℓ and gives results consistent with a previous prediction that relies on a more complex model. Finally, by specifying the model in a way suggested by the work of Kerman *et al.*, it predicts the specific shape of the rising edge of a pulse and the dependence of the rise time on the bias current I_b of the detector. These theoretical predictions hold regardless of detector material, and Chapter 4 explores experimental verifications performed with a detector made from a proprietary amorphous superconductor in a similar class as WSi and MoSi, as well as results from the literature on NbN.

3.2 Qualitative Description of Detection Event

A typical high-detection-efficiency SNSPD consists of a thin ($\sim 5\text{--}10\text{-nm}$ thick) and narrow (width $w \sim 100\text{ nm}$) superconducting film shaped in a meander that matches the optical field mode of the photon source as illustrated in Fig. 3.1(a). Electrically, an SNSPD can be treated using lumped circuit elements coupled to a readout circuit, as shown in Fig. 3.1(b). The detector bias current splits between two pathways to ground: current I_{det} passing through the SNSPD and the signal current I_s passing through the readout circuit load resistor R_L . The SNSPD is treated as a kinetic inductance L_k , connected in series with a parallel combination of a time-dependent resistance, $R_N(t)$ and a switch. The presence of L_k is an effect inherent to superconductors which I discuss in more detail in Section 3.6, but mathematically it behaves identically to an electromagnetic inductance. The resistive part of the nanowire due to photon detection is represented with $R_N(t)$, and the switch in the closed state describes the entire device being superconductive. In the absence of photons, the nanowire is in thermal equilibrium with the substrate at temperature $T_o < T_c$, where T_c is the superconducting critical temperature at zero bias, and the switch is closed ($I_{\text{det}} = I_b$, $I_s = 0$).

A detection event begins with the absorption of a photon by the nanowire with an energy much greater than the superconducting gap energy (Figs. 3.1(a-i)), giving rise to a hot-spot¹. While there continues to be some uncertainty as to the appropriateness of the so-called hot-spot mechanism during the very short time during the initiation of a detection event (Bulaevskii, Graf, and Kogan, 2012; Renema *et al.*, 2014; Casaburi *et al.*, 2015; Engel *et al.*, 2015b; Gaudio *et al.*, 2016), it predicts many experimental observations. The model presented here considers only the dynamics after the hot-spot has already grown across the entire cross-sectional area of the nanowire (Semenov, Goltsman, and Korneev, 2001).

After its initial formation, the hot-spot spreads rapidly due to quasiparticle diffusion and

¹Note that throughout here I use the term hot-spot to refer to the normal region as it grows and shrinks due to Joule heating, in contrast with the localized resistive region present immediately following photon absorption before Joule heating takes place.

FIGURE 3.1: (a) Depiction of an SNSPD meander with incident photons. (a-i) Illustration of a region of the SNSPD near an incident and absorbed photon. (a-ii:v) The progression of hot-spot formation, growth, and decay, which is described using the front velocity v . (b) Lumped-element circuit diagram of an SNSPD and readout. (c) An example SNSPD readout pulse. The vertical gray line marks $t = 0$.

extends across the width of the nanowire (Fig. 3.1(a-ii)), giving rise to a resistive (normal) wire segment with resistance R_{hs} . In the lumped-element circuit model, the switch is now open. The hot-spot continues to grow along the length of the wire due to Joule heating from current I_{det} passing through the normal region, and this growth is characterized by a phase front velocity $v = v(I_{\text{det}})$ at its boundaries (Fig. 3.1(a-iii:iv))². As the resistive region grows, more current is shunted into the readout circuit, resulting in the rising edge of the electrical pulse shown in Fig. 3.1(c). As more current is shunted out of the detector, the hot-spot growth slows and stalls

²This treatment is widely used to describe SNSPD dynamics and is appropriate as long as the thermal healing length L_H is much smaller than the length of the hot-spot. For NbN based detectors, $L_H = 20$ nm, therefore this approximation is valid for all but the very early initial stages of hot-spot development (Yang *et al.*, 2007; Kerman *et al.*, 2009).

at a steady-state current I_{ss} ³.

Shortly after its growth stalls, the hot-spot collapses as cooling to the substrate dominates over Joule heating, as shown in Fig. 3.1(a-v). Once the nanowire returns to the superconducting state, the switch in the lumped-element circuit model is closed and the readout pulse begins to decay exponentially on a time scale governed by the time it takes for the current to return to the inductor (Eq. 3.2 with $R_{hs} = 0$), which has a $1/e$ recovery time of $\tau_r = L_k/R_L$. Note during the decay of the pulse, both L_k and R_L are constant. For high-speed readout circuits, R_L is often $50\ \Omega$, so that τ_r is tens of nanoseconds - much longer than the rise time of I_s .

3.3 Detection Model

The dynamics of the coupled system during a detection event are governed by the interplay between the superconducting nanowire and the readout circuit described using an electro-thermal model given by

$$\frac{dR_{hs}}{dt} = 2\frac{R_{max}}{\ell}v(I_{det}) \quad R_{hs} \geq 0, \quad (3.1)$$

$$L_k \frac{dI_{det}}{dt} + nR_{hs}I_{det} = (I_b - I_{det})R_L, \quad (3.2)$$

which are valid immediately after the initiation of a detection event, with initial conditions $R_{hs}(0) = 0$ and $I_{det}(0) = I_b$. Here, R_{max} is the resistance of the nanowire when its total length ℓ is in the normal state, which occurs when $T > T_c$ at $I_b = 0$ or $I_b \gg I_{sw}$ at $T = 0$, where I_{sw} is the switching current. Thus R_{max}/ℓ is the resistance per unit length of the nanowire and Eq. 3.1 expresses the above description of hot-spot growth. The factor of two accounts for the two fronts that bound a hot-spot. Physically, $R_{hs} > 0$, but the system of equations allows for unphysical solutions arising from the superconductivity transition of the hot-spot. Keeping the solution in the physical domain requires setting $R_{hs} = 0$ when R_{hs} falls below zero in solving

³Some authors call this the retrapping current (Hazra *et al.*, 2010; Smirnov *et al.*, 2016).

the system of equations. Here it is assumed that $v(I_{\text{det}})$ only depends on the present value of I_{det} and not on the size or history of the hot-spot.

Equation 3.2 is Kirchhoff's voltage law for the SNSPD-load resistor loop. The electro-thermal model allows for the possibility that n photons are absorbed by the film simultaneously, as might happen when illuminating the nanowire with a short-duration multi-photon wavepacket. Further, it is assumed that each absorbed photon generates a hot-spot with identical behavior and that they do not overlap spatially, which is appropriate for a small number of hot-spots with typical stalled hot-spot maximum lengths ($\sim 1 \mu\text{m}$) and typical values of ℓ ($\sim 500 \mu\text{m}$) (Yang *et al.*, 2007). The total nanowire resistance is then given by $R_N(t) = nR_{\text{hs}}(t)$. While the hot-spots do not interact directly in the model, they are coupled indirectly through the electro-thermal interaction.

3.4 Characteristic Time and Resistance Scales

3.4.1 Rescaling Technique

Nondimensionalization is a technique where an equation or system of equations is nondimensionalized. It is often used to identify characteristic scales in a dynamical system, and can be particularly useful when working with differential equations because it allows for the extraction of such scales without necessarily solving the system. It is often used synonymously with the term scaling, and I will use this term throughout this thesis to avoid any confusion as to the dimensionality of quantities like photon number. The procedure for rescaling typically involves introducing characteristic scales for each of the independent and dependent variables in the system, and then solving for those variables such that as many terms as possible are close to unity. Typically when a problem involves multiple scales of the same type, a choice must be made as to the most appropriate scaling parameter for the regime of interest, as discussed below.

As a familiar example, we will rescale the equations for a damped harmonic oscillator,

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0, \quad (3.3)$$

$$x(0) = A, \quad (3.4)$$

$$\left. \frac{dx}{dt} \right|_{t=0} = 0 \quad (3.5)$$

where m is the mass, b is the coefficient of damping, k is the spring constant, and A is some initial displacement of the mass. The first step in rescaling this equation is to introduce dimensionless quantities $\tilde{x} = x/\xi$ and $\tilde{t} = t/\tau$, where ξ and τ are the characteristic length- and time-scales respectively. By doing this we arrive at the dimensionless equations

$$\frac{d^2 \tilde{x}}{d\tilde{t}^2} + \frac{b\tau}{m} \frac{d\tilde{x}}{d\tilde{t}} + \frac{k\tau^2}{m} \tilde{x} = 0, \quad (3.6)$$

$$\tilde{x}(0) = A/\xi, \quad (3.7)$$

$$\left. \frac{d\tilde{x}}{d\tilde{t}} \right|_{\tilde{t}=0} = 0. \quad (3.8)$$

The next step is to solve for ξ and τ such that as many terms as possible are near unity. For ξ , this is straightforward and we can see that $\xi = A$. For τ , there are two possibilities

$$\tau_1 = \sqrt{\frac{m}{k}}, \quad (3.9)$$

$$\tau_2 = \frac{m}{b}. \quad (3.10)$$

Here, τ_1 is the familiar time-scale of an undamped harmonic oscillator, and τ_2 is a time-scale associated with the damping of the system. If we were most interested in the oscillatory behavior of the system, it would make the most sense to rescale in terms of τ_1 , and vice versa for the damping behavior. Finally, since the solution for a damped harmonic oscillator is well known,

$$x(t) \propto \text{Re} \left[a_1 e^{((b+\sqrt{b^2-4mk})/2m)t} + a_1 e^{((b-\sqrt{b^2-4mk})/2m)t} \right] \quad (3.11)$$

we can confirm that we have correctly identified the two time-scales. When there is no damping and $b = 0$, $x(t) \propto \cos(\sqrt{k/m} t)$ and we recover τ_1 . When there is no spring and $k = 0$, $x(t) \propto e^{(b/m)t}$ and we recover τ_2 .

3.4.2 Rescaling SNSPD Model

By rescaling Eqs. 3.1 and 3.2, we can explore the effects of various parameters on detector rise times. There are two timescales inherent in this system of equations, τ_r and t_{ch} ; the latter of which is the characteristic timescale for the dynamics related to the rising edge of the readout pulse. The quantity t_{ch} is directly proportional to the rise time of the pulse, but I will refer to it as t_{ch} throughout this thesis since there are many different standards for defining rise time experimentally. It follows that to study the latter regime we rescale Eqs. 3.1 and 3.2 by taking t_{ch} to be the dominant timescale. This is achieved by introducing dimensionless quantities $\tilde{t} = t/t_{ch}$, $\tilde{\tau}_r = \tau_r/t_{ch}$, $\tilde{R}(\tilde{t}) = R_{hs}(t)/R_{ch}$, $\tilde{R}_{max} = R_{max}/R_{ch}$, $\tilde{I}(\tilde{t}) = I_{det}(t)/I_b$, and $\tilde{v}(\tilde{I}) = v(I_{det})/v_b$, where $v_b = v(I_b)$ is the phase front velocity at the start of hot-spot formation. Here, the characteristic resistance R_{ch} represents the maximum resistance attained by a single hot-spot and it and t_{ch} are given by

$$R_{ch} = \sqrt{\frac{4v_b R_{max} L_k}{n\ell}}, \quad (3.12)$$

$$t_{ch} = \frac{2L_k}{nR_{ch}} = \sqrt{\frac{\ell L_k}{nv_b R_{max}}}, \quad (3.13)$$

respectively. Substituting these expressions into Eqs. 3.1 and 3.2 results in the dimensionless set of equations

$$\frac{d\tilde{R}}{d\tilde{t}} = \tilde{v}(\tilde{I}) \quad \tilde{R} \geq 0, \quad (3.14)$$

$$\frac{d\tilde{I}}{d\tilde{t}} + 2\tilde{R}\tilde{I} = \frac{1}{\tilde{\tau}_r}(1 - \tilde{I}). \quad (3.15)$$

In the rescaling of Eqs. 3.12 and 3.13, the dependence on photon number n was the unique choice for eliminating the explicit n dependence in Eqs. 3.1 and 3.2. There is still a “hidden” dependence in that systems with different numbers of photons will have different values for $\tilde{\tau}_r$.

For typical SNSPDs, the turn-on dynamics are much faster than the recovery time τ_r , such that $\tilde{\tau}_r \gg 1$ ($nR_{ch} \gg 2R_L$), and it is possible to neglect the right-hand-side of Eq. 3.15. Assuming operation in this domain, the coupled equations become

$$\frac{d\tilde{R}}{d\tilde{t}} = \tilde{v}(\tilde{I}) \quad \tilde{R} \geq 0, \quad (3.16)$$

$$\frac{d\tilde{I}}{d\tilde{t}} + 2\tilde{R}\tilde{I} = 0, \quad (3.17)$$

with initial conditions $\tilde{I}(\tilde{t} = 0) = 1$ and $\tilde{R}(\tilde{t} = 0) = 0$. $\tilde{v}(\tilde{I})$ plays the role of a driving term for the system of equations. The phase front velocity is taken to be some (perhaps unknown) function of the detector current $v(I)$. However, for all cases having the same bias current I_b , the scaled velocity function $\tilde{v}(\tilde{I})$ is identical – these cases are all described by exactly the same scaled equations above and hence have exactly the same scaled solutions for $\tilde{I}(\tilde{t})$ and $\tilde{R}(\tilde{t})$. In particular, the relative values for the characteristic time-scale for the turn-on dynamics t_{ch} are independent of the specific form of $\tilde{v}$.

Therefore, for a detector operating at a specified I_b , we can directly relate the signal for a multi-photon event to that of a single-photon event by explicitly expressing the solution of Eqs. 3.16 and 3.17 for the scaled current in terms of the actual, physical quantities for the two cases:

$$I_{n=1}(t) = I_n(t/\sqrt{n}). \quad (3.18)$$

Thus, the absorbed photon number n is encoded within the turn-on dynamics, which can be revealed by measuring the 10% - 90% signal rise time or the maximum value of $dI_{s,n}/dt \propto \sqrt{n} dI_{s,1}/dt$. This explains the recent experimental findings of Cahall *et al.*, which are discussed

in more detail in Chapter 4 (Cahall *et al.*, 2017).

3.5 Specifying Phase Front Velocity

More detailed information may be gleaned from Eq. 3.13 by assuming a specific functional form for $\tilde{v}(\tilde{I})$. The physics of a propagating superconducting-normal boundary, traveling at a phase velocity $v(I_{\text{det}})$, for long, narrow superconductors has been studied extensively because of its importance in understanding quenching of superconducting magnets (Gurevich and Mints, 1987). For this analysis, I follow Kerman *et al.* (Kerman *et al.*, 2009) by using an approximate solution to the phase front velocity originally derived by Broom and Rhoderick (Broom and Rhoderick, 1960), which in physical, unscaled units is

$$v(I) = \sqrt{2}v_o \frac{I^2/I_{\text{ss}}^2 - 1}{\sqrt{I^2/I_{\text{ss}}^2 - 1/2}}. \quad (3.19)$$

Here, $v_o = \sqrt{\alpha\kappa/d}/c$ where κ , c , and d are the thermal conductivity, specific heat per unit volume, and thickness of the nanowire, respectively, and α is the heat conductivity coefficient for cooling to the substrate. Note that $v(I_{\text{ss}}) = 0$ from Eq. 3.19, as discussed qualitatively above. An important quantity in studying these problems is the Stekly parameter, which describes the relative magnitudes of Joule heating and cooling to the substrate, $s = 2(I_{\text{sw}}/I_{\text{ss}})^2$ (Gurevich and Mints, 1987; Kerman *et al.*, 2009). For most SNSPDs, s is large⁴. A device is usually biased near I_{sw} , therefore, over most of the rise time, Eq. 3.19 may be approximated as the linear relation $v(I) = \sqrt{2}v_o I/I_{\text{ss}}$, and importantly $v_b = \sqrt{2}v_o I_b/I_{\text{ss}}$. Under these assumptions and using Eq. 3.13, the additional scaling relation emerges: $t_{\text{ch}} \propto 1/\sqrt{I_b}$.

Taking this linear thermal model, Eqs. 3.16 and 3.17 may be solved in terms of elementary functions given by

⁴For the experimental results described in Chapter 4, $s = 300$

$$\tilde{I}(\tilde{t}) = \text{sech}^2(\tilde{t}), \quad (3.20)$$

$$\tilde{R}(\tilde{t}) = \tanh(\tilde{t}), \quad \tilde{t} \geq 0, \quad (3.21)$$

where it is still assumed that the term proportional to $1/\tilde{\tau}_r$ in Eq. 3.15 can be ignored. Note this form of $\tilde{v}$ does not allow for detector reset and hence only describes the SNSPD turn-on dynamics.

The time at which $\tilde{I}_s$ ($\tilde{R}$) reaches a value of $1/2$ is given by $\tilde{t}_{1/2} \sim 0.881$ (~ 0.549), consistent with the scaling behavior discussed above for a generic thermal model. To directly compare this model to experiment, only a knowledge of I_b and two scale parameters is needed. Therefore, for large τ_r and s , Eqs. 3.20 and 3.21 constitute a universal model for SNSPD turn-on dynamics

Comparing the predictions of the exact model of Ref. Kerman *et al.*, 2009 using Eqs. 3.14, 3.15, and 3.19 without approximation, and the universal model given by Eqs. 3.20 and 3.21 reveals very good agreement for the device turn-on dynamics, as seen in Fig. 3.2. Here, I show the temporal evolution of $\tilde{I}_s = 1 - \tilde{I}$ and $\tilde{R}$ for three different operating conditions: 1) one- and 2) three-photon detection events for a typical value of $I_{ss} \ll I_b \lesssim I_{sw}$; and 3) a one-photon event at $I_b \sim 6I_{ss}$, which is a value smaller than that used in typical experiments. For all cases, the universal model agrees well with the exact model up to $\tilde{t} \sim 2$. Beyond this time, a distinct change in slope in the curves for $\tilde{I}$ predicted by the exact model appears due to the detector returning to the superconducting state ($\tilde{R}$ jumps abruptly to zero). For all curves up to $\tilde{t} = 1$, the difference between the exact and universal models is $< 3\%$ for $\tilde{I}$ and $< 4\%$ for $\tilde{R}$, and the small disagreement is greatest for lower I_b , as expected. Thus, the universal model is an excellent tool for understanding typical SNSPD turn-on dynamics.

FIGURE 3.2: Comparison of the exact (Eqs. 3.14 and 3.15) and universal model solutions (Eqs. 3.20 and 3.21, dashed line) for current (a) and resistance (b). Here, the parameters for the exact model are: (blue line) $n = 1$, $I_b = 12.5 \mu\text{A}$; (red line) $n = 3$, $I_b = 12.5 \mu\text{A}$, and (green line) $n = 1$, $I_b = 6 \mu\text{A}$. For all curves, $R_L = 50 \Omega$, $L_k = 0.824 \mu\text{H}$, $v_0 = 60 \text{ pm/ns}$, $R_{\text{sq}} = 461 \Omega$, $w = 70 \text{ nm}$, and $I_{\text{ss}} = 1.04 \mu\text{A}$.

3.6 Kinetic Inductance Considerations

3.6.1 Origin of Kinetic Inductance

In a regular metal, electrons under the influence of an electromagnetic field travel slowly, scattering off each other and contributing to a net drift velocity. In contrast, in a superconductor, electrons form Cooper pairs and no longer scatter. As a result, there is significant inertia associated with their motion; if an applied electromagnetic field suddenly changes direction, there is some lag associated with the time it takes to slow down the Cooper pairs and redirect them. We call this phenomena the kinetic inductance L_K . In thin film superconductors, as well as some high frequency applications, this inductance typically dominates over electromagnetic inductance.

For superconductors, L_K is both temperature and current dependent (Tinkham, 2004). Throughout Chapters 3 and 4, I assume constant temperature; however, I will discuss the temperature dependence in greater detail in Ch. 6. Since I_{det} fluctuates significantly during SNSPD operation, it is worth considering the effect this may have on L_K and verifying that this does not

invalidate the above theoretical predictions.

3.6.2 Effect of Current-Dependent Kinetic Inductance

The kinetic inductance L_k of a superconducting nanowire is a function of the current density in the nanowire. Clem and Kogan (Clem and Kogan, 2012) have shown theoretically that near the superconductor's switching current, L_k can vary by a large ($> 1/2$) fraction from its low current value. Santavicca *et al.* (Santavicca *et al.*, 2016) found experimentally that variations $\leq 10\%$ are typical over the entire range of current values for which their nanowires remained superconducting. Similarly, variations of 8% were observed in a WSi device (Nicolich *et al.*, 2017). In integrating Eq. 3.16 and 3.17, we assume that L_k is constant. This should lead to a small error in pulse shape predictions for the initial part of the detector signal when the nanowire current is large and decreasing and its kinetic inductance is changing. However, this effect is below the experimental sensitivity in this thesis.

By contrast, the scaling relations with photon number and nanowire length are entirely unaffected by any kinetic inductance dependence on current. To see this, we begin by defining the normal state resistance and kinetic inductance per unit length of nanowire, $\mathcal{R} = R_{max}/\ell$ and $\mathcal{L}_k(I) = L_k(I)/\ell$, respectively. Here the kinetic inductance is allowed to depend on the detector current. The model Eqs. 3.1 and 3.2 may be rewritten as

$$\frac{dR_{hs}(t)}{dt} = 2\mathcal{R}v(I_{det}(t)) \quad R_{hs} \geq 0, \quad (3.22)$$

$$\ell\mathcal{L}_k(I_{det}(t))\frac{dI_{det}(t)}{dt} + nR_{hs}(t)I_{det}(t) = 0. \quad (3.23)$$

Consistent with the above treatment, the RHS of Eq. 3.23 is set equal to 0. We define scaled variables τ , r , and i to arrive at the scaling relations

$$t = \sqrt{\frac{2n}{\ell}} \tau, \quad (3.24)$$

$$R_{\text{hs}}(t) = \sqrt{2n\ell} r(\tau), \quad (3.25)$$

$$I_{\text{det}}(t) = i(\tau). \quad (3.26)$$

In the scaled variables, Eqs. 3.22 and 3.23 become

$$\frac{dr(\tau)}{d\tau} = \mathcal{R}v(i(\tau)) \quad 0 \leq R_{\text{hs}}, \quad (3.27)$$

$$\mathcal{L}_k(i(\tau)) \frac{di(\tau)}{d\tau} + r(\tau)i(\tau) = 0. \quad (3.28)$$

The initial conditions for this system are $i(\tau = 0) = I_b$ and $r(\tau = 0) = 0$. The functions $\mathcal{L}_k(i)$ and $v(i)$ are intrinsic functions of the nanowire. Thus, for a specific bias current and nanowire, pulses with different numbers of photons or from detectors of different lengths are described by exactly the same scaled system, even if the kinetic inductance of the detector depends on current. The solution for this scaled system $i(\tau)$, $r(\tau)$ is unique and scaling relations for different rise times for pulses of different photon numbers or from detectors with different lengths are found by expressing the scaled solutions in terms of the physical variables via Eqs. 3.26. For example, for two detectors A and B ,

$$\begin{aligned} I_{\text{det}}^A \left(\sqrt{\frac{n_A}{\ell_A}} t \right) &= i(\tau) = I_{\text{det}}^B \left(\sqrt{\frac{n_B}{\ell_B}} t \right) \\ \Rightarrow I_{\text{det}}^A \left(\sqrt{\frac{n_A \ell_B}{n_B \ell_A}} t \right) &= I_{\text{det}}^B(t). \end{aligned} \quad (3.29)$$

Consequently, the functional shapes of the rising-edges of their pulses is mathematically similar in the sense of Eq. 3.18.

3.7 Experimental Predictions

Before moving on to the experimental realm, it is useful to visualize exactly what we expect to observe. By numerically integrating Eqs. 3.1, 3.2, and 3.19, we can predict the shape of an SNSPD readout pulse for different n , including its rise and fall, as shown in Fig. 3.3. On the timescale of the recovery of the pulse (Fig. 3.3(a)), it is impossible to discern a difference based on n ; however, by focusing on just the rising edge of the pulse (Fig. 3.3(b)) and its derivative (Fig. 3.3(c)), the difference in rise time is clearly visible. In Ch. 4, we will collect many SNSPD readout pulses and bin them based on the maximum of their derivatives. In the absence of noise, this would result in a rather simple histogram with just three δ functions. However, we expect that noise will cause the δ functions to broaden into Gaussians, and how much they broaden will determine the quality of PNR that is possible.

3.8 Conclusion

In conclusion, here we have derived a universal model for the turn-on dynamics of SNSPDs that identifies characteristic time and resistance scales, which can be used to predict the observed detector behavior. Even though there are many seemingly independent device parameters, they contribute to t_{ch} and R_{ch} in a highly dependent manner. Most importantly, this model explains the multi-photon resolution observed recently in SNSPDs. Additionally, it allows for further predictions on the effect of ℓ and I_b on detector rise times, which will be further analyzed in the following chapter.

FIGURE 3.3: Predicted SNSPD readout pulses for $n = 1$ (grey), 2 (blue), and 3 (red) based on numerically integrating Eqs. 3.1, 3.2, and 3.19. (a) Full pulses including their recovery. (b) Rising edges of the pulses. (c) Derivatives of the rising edges of the pulses. For all curves, $I_b = 12.5 \mu\text{A}$, $R_L = 50 \Omega$, $L_k = 0.824 \mu\text{H}$, $v_0 = 60 \text{ pm/ns}$, $R_{\text{sq}} = 461 \Omega$, $w = 70 \text{ nm}$, and $I_{\text{ss}} = 1.04 \mu\text{A}$.

Chapter 4

Experimental Verification of Electro-thermal Model

4.1 Introduction

In this section I provide verification for the model presented in Section 3 by describing experimental evidence as well as by drawing from the literature. First, we demonstrate that superconducting nanowire single-photon detectors (SNSPDs) have inherent photon-number resolving (PNR) capabilities. Then, by exploring the effects of number of absorbed photons and length of the superconducting nanowire, we see that Eq. 3.13 provides an accurate scaling prediction for the rise time of SNSPD readout pulses. Further, we use the universal model described by Eqs. 3.16 and 3.17 to extract further evidence of how the rise time scales with photon number and detector bias current. The experimental data shown in this chapter was collected by Clinton Cahall, while the analysis was done by me under the guidance of Daniel Gauthier and Gregory Lafyatis.

FIGURE 4.1: Schematic of the source setup used in generating multi-photon wavepackets.

4.2 Demonstration of Photon-Number Resolution in SNSPDs

We collect readout pulses from a single-pixel SNSPD illuminated by a pulsed laser source with the goal of identifying optical wavepackets with $n = 1, 2$, or 3 photons. Due to the Poissonian nature of lasers, we cannot know the exact number of photons in a given readout pulse. Instead, we collect many readout pulses and examine them collectively to identify if they follow Poisson statistics.

4.2.1 Experimental Setup

The source consists of a distributed feedback laser (Fitel F0L15DCWC-A82-19340-B) operating at 1550 nm that is intensity modulated via electro-optic modulators (EOSpace). The repetition rate and width of the modulation signal is controlled by a field-programmable gate array (FPGA; Altera Stratix V 5SGXEA7N2F40C2), shown in Fig. 4.1. Two sequential modulators are used to increase the overall extinction ratio between the "ON" and "OFF" states. After creating the pulse with the modulators, it is attenuated to the desired mean photon number per pulse with a variable optical attenuator before traveling to the detector. In the data presented in this thesis, the width of the modulated optical pulse is ~ 80 ps, the repetition rate is 610 kHz, and the mean photon number per pulse is ~ 1.26 .

The detector is a single-pixel meander made of a proprietary amorphous superconducting material from Quantum Opus (*Quantum Opus*), which I refer to throughout this thesis as A45. We operate A45 at 850 mK on the coldfinger of a ^4He sorption refrigerator made by Chase Research Cryogenics (*Chase Research Cryogenics*), which itself is mounted on the cold finger of a closed-cycle ^4He refrigerator housed in a custom-built cryostat.

To resolve changes in the rise-time of the electrical waveform, we use a low-noise, high-bandwidth read-out circuit, which is shown in Fig. 4.2, together with the lump-element depiction of the SNSPD. The cryogenic pre-amplifier is model CITLF3 from Cosmic Microwave Technologies, with a specified analog bandwidth of 10-2000 MHz, a noise temperature of 4 K, and a gain > 30 dB. An additional amplifier (Mini-Circuits ZFL1000-LN, specified bandwidth 0.1-1000 MHz) at room temperature boosts the signal well above the noise-floor of the oscilloscope we use to collect the waveforms (Agilent Infiniium 80404B, 8 GHz analog bandwidth, 40 Gsamples/s).

To mitigate current back-action that affects AC-coupled SNSPD readout circuits (Kerman *et al.*, 2013), we use a passive cross-over network at the input to the CITLF3. This network provides a DC path-to-ground that prevents charging of the input capacitor to the amplifier without degrading the fast rise-time of the waveform (Cahall, Gauthier, and Kim, 2018). Waveforms are recorded at different values for n and I_{bias} using the oscilloscope.

Since the rising edge of an SNSPD readout pulse rises very quickly (< 500 ps), an important factor to consider is whether the readout circuit will have sufficient bandwidth to resolve this effect. I discuss this in further detail in Sec. 4.3.1.

4.2.2 Results

We collect 10k SNSPD readout pulses using the configuration described in the previous section. We then differentiate the waveforms to simplify the process of identifying differences in their rising edges. We expect the maxima of the differentiated readout pulses to scale with $\sqrt{n}$, as described in Section 3.4.2. Fig. 4.3 shows a histogram of the maximum differentiated readout

FIGURE 4.2: Schematic of the experimental setup used for collecting the waveform data. The SNSPD is shown as the typical lumped-element depiction with a kinetic inductance L_k and a hot-spot resistance R_N that is a function of the absorbed photon number n .

pulse height. By fitting a sum of Gaussian functions to the data, we can identify three distinct modes. We expect that if we are truly measuring photon-number, the number of counts in each one of these modes should follow Poisson statistics due to the fact that the source is an attenuated laser. To test this, we repeat the above fitting process with the added constraint that the ratio of areas under the individual Gaussians must obey Poisson statistics. The resulting fits are shown in Fig. 4.4. The reduced chi-square χ_r^2 associated with this fit is ~ 2.4 , which indicates that the data are likely following Poisson statistics. Thus, this provides strong evidence that it is possible to resolve photon number with this technique using only a single-pixel SNSPD.

For a complete understanding of SNSPD PNR, it is important to quantify the quality of the resolution demonstrated here. For an experiment that requires a detector that is capable of distinguishing between single-photon events and multi-photon events, this technique is already quite promising, with $< 0.3\%$ of $n = 1$ events being mistaken for $n > 1$ events, and less than $< 2\%$ of $n > 1$ events being mistaken for $n = 1$ events. Further work is required to attempt to

FIGURE 4.3: Histogram of differentiated peak heights (blue) for $I_b=11 \mu\text{A}$. We fit a sum of Gaussian functions (red) of the form $\sum_{n=1}^3 a_n e^{-((x-b_n)/c_n)^2}$ to the distribution. Resulting individual Gaussians (orange curves) and their centers (gray vertical lines) are also shown. Here, $a_1 = 0.95 \pm 0.01$, $b_1 = 57.38 \pm 0.04$, $c_1 = 4.59 \pm 0.06$, $a_2 = 0.355 \pm 0.009$, $b_2 = 77.7 \pm 0.4$, $c_2 = 8.1 \pm 0.4$, $a_3 = 0.095 \pm 0.009$, $b_3 = 93 \pm 1$, and $c_3 = 8 \pm 2$.

improve the resolution of multi-photon events, and I discuss some possibilities in Chs. 5 and 7.

4.3 Effect of Photon Number on SNSPD Readout Pulses

To test the prediction that $\tau_{\text{rise}} \propto 1/\sqrt{n}$, we pull raw (undifferentiated), representative $n = 1, 2$, and 3 readout pulses from the centers of each corresponding Gaussian fit in Fig. 4.3, and rescale them using the same principle as Eq. 3.18. In Fig. 4.5(a), we show the rising edges of the waveforms collected at the readout. Figure 4.5(b) shows the waveforms after rescaling by $\sqrt{n}$, where it is seen that the waveforms appear to fall on a single curve. This phenomena is more

FIGURE 4.4: Histogram of differentiated peak heights (blue) for $I_b=11 \mu\text{A}$. We fit a sum of Gaussian functions (red) of the form $\sum_{n=1}^3 a_n e^{-((x-b_n)/c_n)^2}$ to the distribution with the constraint that the area under each individual Gaussian (orange curves) obeys Poisson statistics. Here, $a_1 = 0.95 \pm 0.01$, $b_1 = 57.38 \pm 0.04$, $c_1 = 4.54 \pm 0.06$, $a_2 = 0.34 \pm 0.04$, $b_2 = 77.2 \pm 0.6$, $c_2 = 7.9 \pm 0.5$, $a_3 = 0.10 \pm 0.02$, $b_3 = 91 \pm 3$, and $c_3 = 11 \pm 3$.

apparent when examining the derivatives of the waveforms, which are shown in Fig. 4.5(c) and rescaled in (d). Note the noise that is clearly visible in Fig. 4.5(c), which may explain some of the width in the Gaussians of Fig. 4.3.

To determine whether there is a statistical significance between the waveforms shown in 4.5, we perform a one-way ANOVA on the derivative waveforms at $t = 0$ (Navidi, 2007). Typical one-way ANOVA results are reported using an F statistic, which represents the ratio of the variance between groups to the variance within groups as a function of degrees-of-freedom between the groups and total degrees-of-freedom, and a p -value representing significance level, where $p < 0.05$ is considered statistically significant. For the raw waveforms, the results are $F(2,12) = 140, p = 5.6 \times 10^{-9}$, which show a statistically significant difference. Post hoc comparisons using the Tukey HSD test show a significant difference between the $n = 1$ and $n = 2$, $n = 2$ and $n = 3$, and $n = 1$ and $n = 3$ groups. In contrast, for the rescaled waveforms, we find $F(2,12) = 1.7, p = 0.22$, indicating no statistical significance between the waveforms at $t = 0$.

Thus, the difference at $t = 0$ in the derivative curves is statistically significant, while there is no statistical significance between the rescaled derivative curves at $t = 0$. Therefore, rescaling the traces by $\sqrt{n}$ reveals a universal curve independent of photon number. Ringing from the amplifier distorts this effect somewhat beyond $t_N = 0.4$ ns in the rescaled curves.

4.3.1 Effect of the Finite Bandwidth of the Detection System

An important consideration for these measurements is amplifier bandwidth. The SNSPD read-out pulses rise extremely fast (< 500 ps), and will therefore be affected by the limited bandwidth of typical amplifiers. The expected decrease in system gain at high frequencies is dominated by the room temperature amplifier, whose gain we measured to be down by 3 dB at $f_{lp} = 2.4$ GHz. Accordingly, we model the system gain's high frequency roll off by a single-pole low pass filter with a time constant $\tau = 2\pi/f_{lp}$. Here, R and C are chosen to reproduce the amplifier's 3 dB corner frequency. For this model, the "actual" signal, $v_{\text{actual}}(t)$ that would be observed for an

FIGURE 4.5: (a) Rising edges of $n = 1, 2, 3$ traces (gray, blue, red) and (b) with $t_N = t/\sqrt{n}$, as well as their derivatives with respect to time (c) and (d).

infinite bandwidth system is expressed in terms of the measured, filtered signal $v_{\text{measured}}(t)$,

$$v_{\text{actual}}(t) = \tau \frac{dv_{\text{measured}}(t)}{dt} + v_{\text{measured}}(t). \quad (4.1)$$

The derivative signal in Fig. 4.5 shows this effect most clearly. Peak heights are inversely proportional to the pulse rise times, and the finite bandwidth of the system means that the measured heights of these curves will be somewhat smaller than the “actual” heights that result from amplifiers with infinite bandwidth. We mathematically model the “measured” curves as Gaussians whose full width at half maximum are taken from the data shown in the figure. Using Eq. 4.1, we predict that for events of 1, 2, and 3, photons, the “actual” peaks were reduced

by 6%, 11%, and 15%, respectively to give those measured values in the figure. Equivalently, the actual rise times are expected to be shorter than the measured rise times by the same amounts, according to this model of the amplifier system's frequency dependence. This size of an effect is right at the edge of the experimental sensitivity given uncertainties in the exact shape of the pulse rise time distributions and we have not identified it in the data. Future, higher precision work and, especially, an extension of this technique to higher photon-number events, will require careful consideration of this effect.

4.4 Effect of Detector Length on SNSPD Readout Pulses

Recent work by Smirnov *et al.* provides an additional test of the electro-thermal model (Smirnov *et al.*, 2016). They modeled and experimentally studied the effect of detector length on single-photon pulse rise times for SNSPDs. They present a two-temperature (quasiparticle/phonon distributions) model described by five coupled differential equations, which was numerically solved to predict readout signal rise times. They tested their predictions by comparing signals from detectors with different ℓ . The authors argue, qualitatively, that the dependence of rise time on ℓ should be nonlinear for $\ell > 20 \mu\text{m}$ and their model equations confirm this. While we cannot make absolute estimates of detector rise times, we can make a stronger claim of the nature of the nonlinear dependence on ℓ . Detectors differing only in length (and operated at the same bias current) will all be described by Eqs. 3.16 and 3.17. In Eq. 3.13, L_k and R_{max} are proportional to ℓ . Therefore, we predict t_{ch} and hence detector rise time, should scale as $\sqrt{\ell}$.

To draw comparisons to the Smirnov *et al.* work, we digitize their theory and data using the software Plot Digitizer (*Plot Digitizer*). We then fit their model calculations to $g + h\ell^{1/2}$, allowing g and h to be fit parameters, and find $g = 15.1 \pm 0.8 \text{ ps}$ and $h = 15.01 \pm 0.04 \text{ ps}/\mu\text{m}^{1/2}$, with a reduced chi-square statistic of ~ 1 . When calculating the χ_r^2 , we assume that the dominant error is the $\sim 1.6 \text{ ps}$ error introduced by digitizing. The digitization program

works by allowing the user to set the axes and hand-select data points to record their values. We arrived at this digitization error by selecting the same data point multiple times.

The electro-thermal prediction for the scaling of the rise time on ℓ and n does not require a solution to the differential Eqs. 3.1 and 3.2, but is based only on the rescaling of the equations in physical units, Eqs. 3.12 and 3.13. Thus, we conclude that these scaling results are due to the thermoelectric coupling and do not depend on the microscopic physics that defines the functional dependence of v on I_{det} .

FIGURE 4.6: Prediction that $t_{\text{rise}} \propto \sqrt{\ell}$ (red curve) compared with theoretical predictions from Smirnov *et al.* Smirnov *et al.*, 2016 (blue dots), alongside their experimental results for NbN on Si/Si₃N₄ (green diamonds) and Si/SiO₂ (orange squares). We fit their theoretical predictions for values $> 20 \mu\text{m}$ (gray vertical line).

4.5 Application of Universal Model to Experimental Results

To further explore the utility of the universal model, we use it to fit to experimental data to determine t_{ch} for the detector. We record I_s for various values of n with $I_b \sim I_{\text{sw}}$, and for

various values of I_b with $n = 1$.

First, we follow the same procedure described in Sec. 4.3 to identify representative $n = 1$, 2, and 3 pulses. We then fit these representative pulses using the universal model (Eq. 3.20) to determine t_{ch} , as shown in Fig. 4.8. The resulting values are given in Table 4.1. Note that with a value of t_{ch} extracted in this manner for a pulse with known n , combined with a measurement of L_k as described below, it is possible to extract R_{ch} as well.

For varying I_b , we use the same procedure and focus only on the $n = 1$ peak for each value of I_b . Fits to the waveforms can be found in Fig. 4.9 and values of t_{ch} are given in Table 4.1.

The χ_r^2 for these fits ranged from 1.1 to 11.5 for $I_b = 6$ to $12.8 \mu\text{A}$, and 6.6 to 14.1 for $n = 1$ to 3. We hypothesize that the increase of χ_r^2 is due somewhat to the limited bandwidth of and ringing behavior observed in the amplifier. The rising edge for $n = 3$ and $I_b = 12.8 \mu\text{A}$ is somewhat faster than the amplifier rise time, and therefore the fit slightly underestimates the actual data when the pulse first begins to rise at ~ -0.15 ns. Similarly, when the pulse begins to round off at its maximum value at ~ 0.2 ns, the amplifier rings, again causing the fit to somewhat undershoot the actual data. This effect should be more pronounced for higher n and I_b where the rise time is shortest, and this is reflected in the higher χ_r^2 for these fits.

FIGURE 4.7: Experimental values of t_{ch} (blue dots) extracted by fitting with the linear model vs. (a) n with $I_b = 11 \mu\text{A}$ and (b) I_b with $n = 1$, as well as their fits (red curves) to the model $t_{\text{ch}} = A/n^a$ or $t_{\text{ch}} = B/(I_b)^b$ respectively.

n	I_b (μA)	t_{ch} (ns)
1	12.8	0.264 ± 0.007
1	12.5	0.275 ± 0.005
1	12.0	0.276 ± 0.007
1	11.5	0.287 ± 0.006
1	11.0	0.294 ± 0.007
2	11.0	0.212 ± 0.006
3	11.0	0.166 ± 0.006
1	10.5	0.310 ± 0.006
1	10.0	0.323 ± 0.004
1	9.0	0.335 ± 0.004
1	8.0	0.358 ± 0.006
1	7.0	0.409 ± 0.008
1	6.0	0.429 ± 0.007

TABLE 4.1: Values of t_{ch} found by fitting experimental waveforms to the universal model solution.

We then perform additional fitting on these extracted t_{ch} values to test the scaling prediction. The t_{ch} values are shown in Fig. 4.7. Considering only the photon-number data with $I_b = 10.6I_{\text{ss}}$, we expect $t_{\text{ch}} = A/n^a$ with $a = 0.5$. We find $A = 295 \pm 5$ ps and $a = 0.51 \pm 0.03$. Therefore, the scaling is consistent with the universally predicted scaling. From A and other independently-measured parameters discussed in Sec. 4.6, we find $v_0 = 96 \pm 3$ m/s. This value of v_0 is smaller than, but of-the-order-of that found for NbN-based SNSPDs (Berggren *et al.*, 2018), and is consistent with the slower turn-on and turn-off dynamics of SNSPDs based on the amorphous superconducting thin film considered here.

In contrast, considering only the bias-current data with $n = 1$, we expect $t_{\text{ch}} = B/(I_b)^b$ with $b = 0.5$; however, we find $B = 1.35 \pm 0.08$ ps $\cdot A^{1/2}$ and $b = 0.63 \pm 0.03$. From B , we predict $v_0 = 50 \pm 5$ m/s. We also simultaneously fit all data using $t_{\text{ch}} = C/n^c(I_b)^d$ and find $C = 1.36 \pm 0.07$ ps $\cdot A^{1/2}$, $c = 0.52 \pm 0.03$, $d = 0.63 \pm 0.02$, and $v_0 = 50 \pm 5$ m/s. In both cases, the dependence of I_b on t_{ch} is inconsistent with the predicted value. The inconsistency may be due to some physics not captured in the model proposed in Ref. Kerman *et al.*, 2009. It may also be due to high measured dark counts (discussed below) for higher I_b measured in this detector,

FIGURE 4.8: Universal model fits (red solid line) for $n = 1, 2$, and 3 (blue points).

FIGURE 4.9: Universal model fits (red solid line) for different I_b (blue points).

FIGURE 4.10: Measurement schematic (a) and resulting IV curve (b) for the Quantum Opus detector. There is a $1\text{ M}\Omega$ series resistor R_{bias} between the voltage source and the detector, resulting in a slope of $1\text{ M}\Omega$ during the superconducting portion of the IV curve. The normal resistance is about $6.53\text{ M}\Omega$ including R_{bias} . The bias-T is a Mini-Circuits ZFBT-6GW and the low pass filter (LPF) is a Mini-Circuits SLP-1.9+.

which may skew t_{ch} to lower values at higher I_b . Finally, it may be related to distortions in the waveform caused by the amplifier as discussed above.

4.6 Determining Detector Parameters

4.6.1 Steady State Current

The steady-state current I_{ss} is defined as the current when the Joule heating and cooling to the substrate are balanced such that $v(I_{\text{ss}}) = 0$. We determine I_{ss} by performing a DC current-voltage (IV) measurement as shown in Fig. 4.10. We start with a high voltage such that the whole meander is in the normal state. We gradually lower the voltage, and observe a plateau region starting at around 6 V and ending around 1 V. In this region, sections of the nanowire are beginning to switch into the superconducting state, causing the resistance to drop. In turn, the current maintains a constant value, I_{ss} . We fit a line to the IV curve from 4-6V and find the vertical offset is $I_{\text{ss}} = 1.042 \pm 0.005\text{ }\mu\text{A}$.

FIGURE 4.11: Measurement schematic (a) and resulting SNSPD pulse (b). We fit Eq. 4.2 (red) to the falling edge (light blue region) and extract L_k . The low pass filter (LPF) is a Mini-Circuits SLP-1.9+.

4.6.2 Kinetic Inductance

To determine L_k , we model the electrical properties of the SNSPD during the falling edge of the readout pulse and fit to experimental waveforms. We place the detector in parallel with two room-temperature amplifiers using a bias-T (Mini-Circuits ZFBT-6GW). Typically, it is assumed that the fall time for I_s is given by $\tau_r = L_k/R_L$. In this case, we modify this assumption to also allow for timing variations caused by the AC-coupling capacitor. We use the model

$$I_s(t) = e^{(R_L/2L_k)(t-t_o)} (ae^{\sqrt{(R_L/2L_k)^2 - 1/(L_k C)}(t-t_o)} + be^{-\sqrt{(R_L/2L_k)^2 - 1/(L_k C)}(t-t_o)}) \quad (4.2)$$

where $R_L = 50 \, \Omega$ and a , b , t_o , L_k , and C are left as free parameters. (C is not specified by the bias-T manufacturer, so we allow it to be a fit parameter.) We fit Eq. 4.2 to the pulse shown in Fig. 4.11 from $t \sim 15$ to $t = 250$ ns. We find $a = -0.059 \pm 0.004$ V, $b = 1.53 \pm 0.01$ V, $t_o = -1.2722 \pm 0.0004$ ns, $C = 3.2 \pm 0.2$ nF, and $L_k = 824 \pm 4$ nH. The fit has $\chi_r^2 \sim 1.6$.

FIGURE 4.12: The detector dark count rate (points with solid line) varies from a few counts per second (cps) for low I_b , up to $> 8,000$ cps for high I_b . The total count rate (squares with dotted line) is also shown. (Note measured results are given by plot points – the lines are just guides for the eye.)

4.6.3 Dark Counts

The dark count rate (DCR) as well as the total count rate (TCR) as a function of I_b for the detector is shown in Fig. 4.12. At higher I_b ($\gtrsim 11 \mu\text{A}$), the DCR grows to within two orders-of-magnitude of the TCR.

Careful analysis of the measured dark counts reveals an interesting phenomenon, which is not accounted for by the model. We collect dark count pulses, differentiate their rising edges, and record their maximum derivatives in the same way as above. We assume that we do not obtain any multi-photon dark count events and so the resulting distribution represents only $n = 1$ counts. We compare the resulting distribution to the multi-photon distribution and find

FIGURE 4.13: Comparison of multi-photon absorption data (blue) with dark count rates (red) at $I_b = 12.8\mu\text{A}$. Both distributions are normalized such that the area under their curves are unity, then the dark counts are further scaled by the ratio of the DCR over the overall count rate during data collection. Gaussians are fit to the $n = 1$ peak and their centers are shown as vertical lines for both source (solid blue) and dark (dotted red) counts.

that the $n = 1$ dark counts are shifted to lower differentiated peak height (longer rise time) as shown in Fig. 4.13. A possible explanation for this shift is that the dark count photons are likely much longer wavelength than the source photons Shibata *et al.*, 2015; however, there is no mechanism in the model to account for this proposed difference. This is an important issue that warrants further exploration.

4.7 Conclusion

In this chapter, we have explored a demonstration of photon-number resolution in SNSPDs, as well as verified predictions for how the rising edge of an SNSPD readout pulse depends on photon-number and detector length. Additionally, using the universal model derived in the previous Chapter, we are able to fit to SNSPD readout pulses and extract further support that the rise time scales as $1/\sqrt{n}$ and $1/\sqrt{I_{\text{bias}}}$, although more work is needed to understand the latter relationship in greater detail. Finally, we use the information extracted with the universal model combined with simple additional measurements to make predictions about the characteristic superconducting-normal front velocity in the nanowire and find results consistent with the literature.

Chapter 5

Ramifications of Spatial Variations in SNSPD Nanowire Parameters

5.1 Introduction

In this chapter, I dive deeper into understanding the multi-photon resolution capabilities presented in the preceding chapters. In particular, I show that the distribution of $n = 1, 2$, or 3 photon measurements we measured is noisier than we would reasonably expect. Then, I discuss potential origins of this additional noise, including an anomalous below- T_C resistance in the SNSPDs. The work presented in this section was completed by me under the guidance of Daniel Gauthier and Gregory Lafyatis.

5.2 Expected Noise for SNSPD PNR

In Ch. 4, I presented experimental evidence that superconducting nanowire single-photon detectors (SNSPDs) are inherently capable of photon-number resolution (PNR). In particular, Fig. 4.3 shows that the SNSPD is producing different readout signals based on the number of photons it absorbs. We can accurately predict the magnitude of the differentiated SNSPD readout pulse based on the theory presented in Ch. 3, and the number of counts we measure for $n = 1$,

2, or 3 photons is consistent with what we would expect based on Poissonian statistics. However, an extremely important factor in the usefulness of PNR is how well the different photon number events can be distinguished from each other. In other words, it is useful to understand the width of the Gaussian function shown in Fig. 4.3, and ideally identify ways to improve the PNR by narrowing those distributions.

To get a better sense for the predicted SNSPD PNR noise, we convert the noise in the SNSPD readout voltage signal into an associated noise in the derivative of the signal, as is used in Fig. 4.3. We start by approximating the derivative as

$$\frac{dV}{dt} \approx \frac{(V_2 + \sigma_2) - (V_1 + \sigma_1)}{\Delta t}. \quad (5.1)$$

Here, σ_1 and σ_2 are uncorrelated random variables drawn from some distribution $\Gamma(\sigma)$, which makes $\frac{dV}{dt}$ a random variable itself. To calculate the uncertainty associated with $\frac{dV}{dt}$, we need to find the uncertainty associated with the probability distribution for the new random variable, $(\sigma_2 - \sigma_1)/\Delta t$. The probability distribution for a difference of random variables is given by

$$P_{(\sigma_2 - \sigma_1)/\Delta t}(\sigma) = \int_{-\inf}^{\inf} \Gamma(-\sigma') \Gamma(\sigma - \sigma') d\sigma'. \quad (5.2)$$

For the case where the noise is Gaussian with standard deviation σ , this becomes

$$P_{(\sigma_2 - \sigma_1)/\Delta t}(\sigma) = \frac{1}{\sqrt{2\pi}(\sqrt{2}\sigma)} e^{-\sigma^2 / (2(\sqrt{2}\sigma/\Delta t)^2)}. \quad (5.3)$$

Therefore, the noise associated with the derivative is $\sqrt{2}\sigma/\Delta t$, which works out to be 0.6 V/ns. Fig. 5.1 shows what we would expect the differentiated peak height distribution to be based on this noise figure. We predict a set of Gaussians that are much more narrow than what was actually measured in the experiment. If we could achieve distributions as narrow as we predicted, this would provide an enormous improvement in the SNSPD PNR. Thus, it is important that we explore potential reasons that the measured SNSPD PNR is significantly worse than expected.

FIGURE 5.1: Comparison of expected (grey dashed) vs. measured (blue) SNSPD PNR distribution. Also shown are Gaussians fit to each node of the distribution (orange) and their sum (red) as previously shown in Fig. 4.3

5.2.1 Potential Causes of Increased Noise in SNSPD PNR

The broader measured Gaussians than expected in Fig. 5.1 practically mean that we are observing a wider range of SNSPD readout signal rise times than anticipated. One potential cause of this could be broadening of the output readout pulse based on the transmission line properties of the SNSPD meander (Santavicca *et al.*, 2016). Santavicca *et al.* showed that SNSPDs have a frequency-dependent impedance, which could conceivably cause broadening of the readout pulse that is based on where in the nanowire a photon is absorbed. More work is required to better understand this possibility.

For the remainder of this thesis I will focus on another potential cause of increased SNSPD PNR, which is that there is some parameter in t_{ch} (3.13) that is fluctuating unexpectedly, perhaps as a result of spatial variations in some characteristic of the nanowire. Therefore, the rise time of the SNSPD readout pulse depends somewhat on where in the nanowire the photon is absorbed.

To explore this idea further, we begin by examining 3.13, which I have reproduced here

$$t_{\text{ch}} = \sqrt{\frac{\ell L_k}{n v_b R_{\text{max}}}}. \quad (5.4)$$

Following the thermal model used in Sec. 3.5 and assuming $I_b \gg I_{\text{ss}}$,

$$v_b \approx \sqrt{2} v_0 I_b / I_{\text{ss}}. \quad (5.5)$$

Finally, $I_{\text{ss}} = 2I_{\text{sw}} / \sqrt{\psi}$, where ψ is the Stelky parameter. Substituting these into 5.4 gives us a new approximate expression for t_{ch} ,

$$t_{\text{ch}} = \sqrt{\frac{\sqrt{2} \ell L_k I_{\text{sw}}}{n v_0 I_b R_{\text{max}} \sqrt{\psi}}}. \quad (5.6)$$

The quantities ℓ , L_k , n , I_b , and R_{max} are all global quantities that are defined or affect the entire

nanowire in the same way, so they cannot contribute to a broadening in the expected range of t_{ch} . In contrast, the switching current I_{sw} , the characteristic normal-superconducting boundary velocity v_0 , and the ratio of the characteristic heat generation in the normal state to the heat transfer (Stekly parameter) ψ , all could conceivably vary along the length of the wire for a variety of reasons. For example, variations in the nanowire width due to imperfect fabrication could cause fluctuations in these quantities.

The dynamics of thin, narrow, current-biased superconductors with spatially varying properties have been studied, in particular in the context of variations in width (Gurevich and Mints, 1987). In the case of a short superconducting microbridge sandwiched between regions of wider superconductor, the spatial variation in width can allow for resistive hot-spots to form and remain stable within the narrow regions of the bridge (Skocpol, Beasley, and Tinkham, 1974). Although the SNSPDs explored here are unlikely to have width variations on the scale explored in Skocpol *et al.*, it is possible that similar variations, for example in I_{sw} , could also result in localized hot-spots that are stable. In this case, we would expect to see some below- T_C resistance in SNSPDs, which has not previously been observed. In Ch. 6, I attempt to explore this concept further as well as attempt to measure such a below T_C resistance experimentally.

5.3 Conclusion

In this section, I showed that the SNSPD PNR resolution we achieved experimentally was worse than predicted due to some unidentified source of noise in the rise times of the SNSPD readout pulse. I hypothesized that this could be related to recently identified properties of impure superconductors, and that if this is true we would expect to see some below- T_C resistance in the SNSPDs. In the following chapter, I attempt to measure R_n experimentally.

Chapter 6

Below- T_C Resistance in SNSPDs

6.1 Introduction

In the previous chapter, I hypothesized that unexpectedly poor superconducting nanowire single-photon detector (SNSPD) photon-number resolution (PNR) may be related to some anomalous below- T_C resistance in the detector. In this chapter, I attempt to measure this resistance using a lock-in amplifier and confirm its existence. Based on these results, I discuss a simple theoretical basis to explain below- T_C resistance in SNSPDs, as well as present a model to understand the effect that an additional resistance could have on the detector dynamics. The work presented in this section was done entirely by me under the guidance of Daniel Gauthier and Gregory Lafyatis.

6.2 SNSPD Resistance Measurements Using a Lock-in Amplifier

We begin by attempting to measure below- T_C resistance using a lock-in amplifier. Lock-in amplifiers work by outputting a sinusoidal signal at a specified frequency, which is sent through the sample before returning to the lock-in. By comparing the output signal to what it receives, the lock-in is able to measure a magnitude and phase associated with the signal, even in an extremely noisy environment.

Since we hope to measure the below- T_C resistance of the SNSPD, the detector must remain nominally superconducting throughout the measurement. Practically, we are measuring a very small resistance ($\mathcal{O}(50\Omega)$), typically requiring a large current for accurate measurement, within a superconducting material which will transition to the normal state at only very small currents ($\mathcal{O}(10\mu A)$). A lock-in amplifier remedies this problem by allowing us to send a very small amount of current through the device, while still accurately measuring the magnitude and phase of the voltage across the detector. With this information we can calculate the complex impedance of the SNSPD and specifically examine the below- T_C resistance.

It is worth noting that this constitutes a two-wire measurement. Often, it is preferred to use a four-wire configuration for precision resistance measurements because this eliminates any lead and contact resistance in the measurement. In this case, it is reasonable to use a two-wire configuration because that is how the detectors are operated conventionally.

6.2.1 Experimental Setup and Procedure

The experimental setup for this measurement is relatively simple and straightforward. The lock-in (SRS 850 DSP) emits a voltage signal V_0 with an amplitude of 10 mV RMS and a variable frequency f , as shown in Fig. 6.1. We place a 100 k Ω resistor before the detector to keep the current flowing through the detector below the superconducting switching current. The lock-in measures the complex voltage across the detector, comprised of L_K and R , in parallel with the capacitance C of the coaxial cables used. We are using around 12 feet of coaxial cable, so we would expect $C \sim 300$ pF based on the RG-58 standard.

To vary the detector temperature, we exploit the cycling dynamics of the Chase 1K Cryocooler (see Appendix A for more details). Cycling this cryocooler allows us to sweep the SNSPD temperature down and back up, varying between 880 mK and 8 K. We record the signal measured by the lock-in every ~ 45 s as we sweep through temperature. Because the detector undergoes a huge change in resistance as it switches between the superconducting state and

FIGURE 6.1: Measurement setup for measuring SNSPD resistance using a lock-in amplifier. Not picture is the lock-in load resistor which is specified at $100\text{ M}\Omega$ and therefore has a negligible effect on V .

the normal state, we allow the lock-in to automatically adjust the gain for each collected data point. We repeat this process for $f = 10$ kHz, 50 kHz, and 100 kHz.

Using this technique, we measure the complex impedance of two different SNSPDs. The first is from Quantum Opus and is referred to as A45 throughout this thesis. A45 is made of a proprietary amorphous material. The second is a WSi detector that was fabricated at the National Institute of Standards and Technology, which I refer to as N1 (*National Institute of Standards and Technology*).

6.2.2 Initial Results

The voltage and phase measured by the lock-in is shown in Fig. 6.2. We can see that A45 undergoes a superconducting transition at ~ 5 K, while N1 transitions closer to ~ 3 K. This is expected since the detectors are made from different materials. N1 is known to be WSi and its transition temperature is consistent with other WSi SNSPDs (Verma *et al.*, 2014). At high temperatures, above T_C , we expect the resistance of the detector to be very large, $\mathcal{O}(100k\Omega)$ or greater. In contrast, the impedance given by $1/(i\omega C)$ will be vary between ~ 5 -500 $k\Omega$ as f changes. Therefore, we expect that the voltage measured by the lock-in when the detector is normal will change with f , which is reflected in the data shown in Fig. 6.2.

A closer look at the below- T_C voltage magnitude and phase is shown in Fig. 6.3. Most striking is the fact that the voltage magnitude changes as a function of below- T_C temperature for both detectors. We do not expect C to have any temperature dependence, so this dependence must be due to either L_K or some anomalous R_a , as introduced in Ch. 5. In SNSPDs, L_k is typically on-the-order-of 1 μH at operating temperatures. L_k is known to be temperature dependent, and groups have shown it grows up to 3 times as detector temperature increases up to T_C (Tinkham, 2004; Santavicca *et al.*, 2016). The smallest voltage measured by the lock-in is seen with A45 and is about 3 μV . For this to be entirely inductive, L_k would need to be almost 50 μH , which is well beyond what is seen in SNSPDs. Therefore, it appears that the temperature-dependent impedance that is causing these changes in V is resistive.

FIGURE 6.2: (a) Voltage and (b) phase measured by the lock-in amplifier using 10, 50, and 100 kHz driving frequency for detectors A45 (blue, red, green respectively) and N1 (purple, pink, orange respectively).

6.2.3 Circuit Model

The complex voltage associated with V in the circuit shown in Fig. 6.1 is given by

$$V = \frac{V_0 Z_{\text{eff}}}{R_b + Z_{\text{eff}}}, \quad (6.1)$$

where Z_{eff} is the effective impedance associated with the SNSPD in parallel with C . When the temperature is below T_C , $R_b \gg Z_{\text{eff}}$, so it is reasonable to use the approximation $V \approx V_0 Z_{\text{eff}} / R_b$. Further, based on the analysis in the previous section, we assume L_k is small enough to be neglected here, which results in a complex voltage given by

$$|V| = \frac{V_0 R}{R_b \sqrt{1 + (\omega C R)^2}}, \quad (6.2)$$

$$\phi = \tan^{-1}(-\omega C R), \quad (6.3)$$

where $\omega = 2\pi f$. With this model, we can extract the value of R based on the the complex voltage measured by the lock-in amplifier.

FIGURE 6.3: Voltage and phase measured by the lock-in amplifier at 10, 50, and 100 kHz driving frequency for detectors (a-b) A45 (blue, red, green respectively) and (c-d) N1 (purple, pink, orange respectively) at low temperatures (below 2.5 K).

FIGURE 6.4: Measured $|V_C|$ as a function of frequency (blue dots) fit with Eq. 6.4 (red line).

6.2.4 Measuring Capacitance

Before extracting the below- T_C resistance, it is important to carefully measure C . To do this, we remove the detector from the circuit shown in Fig. 6.1, while leaving all of the coax that runs to the detector intact. This results in a voltage magnitude given by

$$|V_C| = \frac{V_0}{\sqrt{1 + (\omega C R_b)^2}}. \quad (6.4)$$

We then measure $|V_C|$ as a function of f , and fit the model to this data with C as a fit parameter, as shown in Fig. 6.4. This results in $C_{A45} = 293.5 \pm 0.7$ pF and $C_{N1} = 295.4 \pm 0.7$ pF.

6.2.5 Below- T_C Resistance Results

To extract R from the complex voltage data, we fit the magnitude of the voltage $|V(2\pi f, T)|$ for varying f at specified values of T , allowing R to be a fit parameter. Data is recorded for each f during different cooling cycles, which results in measurements at different absolute values of

FIGURE 6.5: Interpolation (red curve) of measured $|V(2\pi f, T)|$ (blue dots) for $f = 50$ kHz for A45 as a function of temperature.

T . To be able to compare $|V|$ at the same, specific values of T , we interpolate the $|V(2\pi f, T)|$ data for each of the three f . An example of this interpolation is shown in Fig. 6.5.

At each value of T , we fit Eq. 6.2 to the measured value of $|V|$ to extract R . Because we swept through T at three different f , this results in three values of $|V|$ for each T . The resulting $R(T)$ is shown in Fig. 6.6. Note, in principle we could also fit to the phase data using Eq. 6.3, but that data was more noisy overall. However, based on the R extracted from fitting to $|V|$, we can predict ϕ , as shown in Fig. 6.7. For A45, the agreement between the prediction and the experiment is excellent for $f = 10$ kHz, with a $\chi_r^2 \sim 0.95$. This agreement is less strong for $f = 50$ kHz and 100 kHz, $\chi_r^2 \sim 2.2$ and 7.0 respectively. This discrepancy may be due to a phase offset in the A45 measurement that is more pronounced at higher frequencies. For N1, the agreement is once again good for $f = 10$ kHz, with $\chi_r^2 \sim 0.9$. Again, the agreement is worse at higher frequencies, with $\chi_r^2 \sim 3.3$ and 13.0 for $f = 50$ kHz and 100 kHz respectively.

It is clear from these results that there appears to be a significant temperature-dependent

FIGURE 6.6: Measured resistance for (a-b) A45 and (c-d) N1 as a function of temperature.

FIGURE 6.7: Predicted phase based on measured resistance (curves) vs. measured phase (points) at 10, 50, and 100 kHz for (a) A45 (blue, red, green respectively) and (b) N1 (purple, pink, orange respectively) as a function of temperature.

below- T_C resistance in both A45 and N1. Since these detectors are made from different amorphous materials and were fabricated in completely different facilities, this suggests that below- T_C resistance may be a feature of amorphous SNSPDs, or perhaps even SNSPDs generally. The resistance for A45 appears to be $\sim 40 \Omega$ at the lowest, which is consistent with a known 50Ω series resistor that was integrated onto its chip by the manufacturer. The N1 data has a higher base resistance than expected, which might suggest an error in the measurement. However, the overall trend of a temperature-dependent resistance is likely real and significant.

6.3 Theoretical Basis for a Below- T_C Resistance in SNSPDs

Measuring a below- T_C resistance in an SNSPD is a very unexpected result, and it may initially be surprising that this has not been observed previously. Most groups working with SNSPDs work at a single, stable temperature, which is typically much lower than the T_C of the device. For example, groups who work with similar detectors to A45 and N1 typically operate at temperatures around 1 K, where we measured a relatively small resistance on-the-order of 10s of Ω . This represents a very small portion of the detector, $\sim 0.01\%$ of its length, or about 100

nm. Based on this, it seems likely that the detector would still function, which I explore further in Sec. 6.4. Additionally, even if a resistance measurement were performed on the detector, it would be tempting to explain away a small resistance at a single temperature as pertaining to another component in the system or as some kind of contact resistance. However, the fact that the resistance presented here is temperature dependent and present across multiple detectors suggests that this may be a real effect relating to the superconducting material.

Here, I would like to lay the groundwork for a theoretical model for a below- T_C resistance in SNSPDs. First, we imagine that this resistance is due to small regions of the nanowire persisting in a resistive state. These regions have to be stable, meaning they do not change in size, unlike the hot-spots caused by photon absorption as discussed in Ch. 3. Based loosely on the theory discussed previously, this means the Joule heating caused by current traveling through the resistive region is exactly balanced by the heat escaping into the substrate of the SNSPD. The number of these small resistive regions is dependent on the temperature of the detector: as the temperature drops, some of these small sections become superconducting and the overall resistance decreases. A reasonable hypothesis is that variations in some detector parameter are causing these regions to exist as well as decreasing the overall SNSPD PNR, as discussed in Ch. 5. An intuitive candidate is the switching current I_{sw} . If I_{sw} is fluctuating along the length of the nanowire x , there may be some regions where the localized $I_{sw}(x) < I_b$, which will cause a localized normal region. In contrast, in the majority of the nanowire that is superconducting, photons can still be detected, although fluctuations in I_{sw} will cause variations in the rise time of the readout pulse, deteriorating the PNR.

Stable normal regions within a superconducting nanowire have been posited before (Skocpol, Beasley, and Tinkham, 1974; Gurevich and Mints, 1987). In their review paper, Gurevich and Mints explored the dynamics of a superconducting-normal wavefront in a long, thin superconducting wire. They found that small normal regions could persist as described above and shown in Fig. 6.8(a), as long as the device was operated at a stable temperature. (Under normal operation and at base temperature, the cryostat used in this thesis maintains a temperature

FIGURE 6.8: Examples of stable hot-spot regions (a) in a long, thin nanowire and (b) at the interface between a microbridge and a bulk superconductor. Figure based on Gurevich *et al.* (Gurevich and Mints, 1987).

stability of < 1 mK.) Further, they describe that these hot-spots are more likely to form in a place where the wire abruptly changes width or shape, such as where the wire connects to a piece of bulk superconductor, as shown in Fig. 6.8(b). It is unlikely that the SNSPDs studied here experience large width fluctuations ($> 10\%$ of the nanowire width) based on the typical resolution of electron-beam lithography used in fabrication (Chen, 2015); however, these detectors are composed of alloys and may experience significant variations in material parameters. For example, in MoSi, a similar amorphous superconducting material to those studied here, the composition of the alloy significantly affects T_C as shown in Fig. 6.9 (Kubo, 1988; Bosworth *et al.*, 2015). It is reasonable to imagine that a similar effect occurs with I_{sw} and that the alloy composition in these detectors could vary spatially, which could explain both the below- T_C resistance and the poor SNSPD PNR as described above.

The elements presented here are the building blocks of a model that might explain the below- T_C resistance measured. However, it is important to note that this is only one possible explanation. Alternatively, recent work suggests that the superconductivity of materials used in SNSPD fabrication is characterized by islands of superconducting material surrounded by regions of normal material, which have interesting characteristics including a nonzero resistance that persists below- T_C (Durkin *et al.*, 2020). Overall, more work is required to better understand and characterize this effect.

FIGURE 6.9: Critical temperature of MoSi thin films as a function of Si composition. 50 nm (square points) and 200 nm (diamond points) data taken from Kubo (Kubo, 1988). 100 nm (circular points) taken from Bosworth *et al.* (Bosworth *et al.*, 2015).

6.4 Ramifications of Below- T_C Resistance on Detector Functionality

It is important to understand how a below- T_C resistance will affect the SNSPD dynamics and if the detectors will still function qualitatively. To explore this, we begin by modeling the SNSPD in a similar way as is described in Ch. 3. Electrically, the SNSPD lumped-element model is the same except for an additional anomalous series resistance in the detector R_a , that is constant and does not switch into a superconducting state. In this configuration, R_a simply causes less of I_b to travel through the SNSPD, decreasing the relative readout signal during a detection event. This is easily remedied by simply increasing I_b until the detector receives a high enough bias to create a larger signal.

However, many SNSPDs are operated with an AC coupled readout as shown in Fig. 6.10. It is important to explore whether in this configuration an anomalous resistance will cause qualitative differences in detector behavior. Electrically, we must adjust the model of Ch. 3 to

FIGURE 6.10: Lumped-element circuit model of SNSPD with anomalous below- T_C resistance R_a and AC coupled readout.

include R_a and the capacitance associated with the AC coupled readout C , and we allow $n = 1$. Thermally, we use the specified thermal model of Sec. 3.5, which yields

$$\frac{dR_{\text{hs}}}{dt} = 2\sqrt{2} \frac{R_{\text{max}} v_o}{\ell} \frac{I_{\text{det}}^2 / I_{\text{ss}}^2 - 1}{\sqrt{I_{\text{det}}^2 / I_{\text{ss}}^2 - 1/2}} \quad R_{\text{hs}} \geq 0, \quad (6.5)$$

$$L_k \frac{d^2 I_{\text{det}}}{dt^2} = (I_b - I_{\text{det}}) / C - \frac{dI_{\text{det}}}{dt} (R_{\text{hs}} + R_a + R_L) - I_{\text{det}} \frac{dR_{\text{hs}}}{dt}. \quad (6.6)$$

with $R_{\text{hs}}(0) = 0$, $I_{\text{det}}(0) = I_b$, and $\frac{dI_{\text{det}}}{dt}|_{t=0} = 0$.

The result of numerically integrating this system of equations for different values of R_a is shown in Fig. 6.11. For small values of R_a , there is only a subtle change in the hot-spot resistance and the readout pulse has a decreased recovery time. As R_a is increased, the hot-spot resistance and the signal current do not return to zero, and the detector is unable to recover into the superconducting state. This is a phenomena called latching that is well documented

FIGURE 6.11: (a) Hot-spot resistance and (b) readout current for a detection event in an AC-coupled SNSPD with $R_a = 0$ (grey dashed), 100 (red), and 200 (blue) Ω . Here, $R_L = 50 \Omega$, the AC coupling capacitor C is 400 pF, $I_b = 10 \mu\text{A}$, and the SNSPD has $L_k = 1 \mu\text{H}$, $v_0 = 50 \text{ m/s}$, $I_{ss} = 1.04 \mu\text{A}$, normal sheet resistance $R_{sheet} = 461 \Omega$, and width $w = 70 \text{ nm}$.

to occur when increasing the load resistor R_L (Yang *et al.*, 2007; Kerman *et al.*, 2013). Indeed, a closer look at Eq. 6.6 shows that for an AC coupled SNSPD, an anomalous resistance R_a in series with the detector will affect the circuit dynamics in exactly the same way as increasing R_L .

Therefore, a below- T_C resistance could significantly impact detector performance by causing the detector to latch and not function correctly at expected temperatures. If the anomalous resistance were itself temperature dependent, we would expect that the SNSPD would require a lower operating temperature than anticipated. There is wide interest in increasing the operating temperature of SNSPDs to reduce cryogenic costs, and thus it is important to understand and characterize this effect fully.

Based on this modeling, we expect the detectors will become insensitive to photons due to latching at a much lower temperature than T_C . To examine this, we probe A45 with 1550 nm light at varying temperatures to see where it stops producing readout pulses. We operate A45 in the configuration shown Fig. 3.1(b), with a CITLF3 cryogenic amplifier as the leading

FIGURE 6.12: A45 readout pulses for various operating temperatures with $I_b = 0.9I_{\text{sw}}$. The detector stops reading out pulses above $T = 2.24$ K.

amplifier. At each temperature T_0 , we measure the switching current $I_{\text{sw}}(T_0)$ by increasing the current until the entire detector transitions to the normal state. We then bias the detector at $0.9I_{\text{sw}}(T_0)$ and observe whether the detector is reading out events. Fig. 6.12 shows the readout for various operating temperatures. A45 stops reading out pulses above $T = 2.25$ K. This corresponds to a resistance of $\sim 150 \Omega$, which is consistent with predictions from Sec. 6.4. This agreement is interesting; however, repeating these SNSPD-sensitivity measurements at $0.7I_{\text{sw}}(T_0)$ showed that the detector remained functional up to 3.3 K. More work is needed to fully understand these results.

6.5 Conclusion

In this section, I have described a measurement scheme for measuring the impedance of an SNSPD using a lock-in amplifier. I derived a model for the complex voltage across the SNSPD assuming this impedance is small, as when the detector is at a temperature below $-T_C$. From the lock-in measurements, we can extract values for R and show that it is nonzero and increases with T , which has not been previously demonstrated in SNSPDs. Based on this, I posited a simple theoretical basis that would allow for a below- T_C resistance. Finally, I modeled how an anomalous below- T_C resistance would affect detector dynamics, and found that it could cause detectors to stop functioning and latch in unexpected temperature ranges, which could have wide-reaching implications for the operation of SNSPDs. This effect is experimentally demonstrated in detector A45. These results challenge the current understanding of SNSPD physics and warrant further exploration.

Chapter 7

Conclusions and Future Research

Through quantum computing, quantum modeling, and quantum cryptography, quantum information science promises to deliver high quality, secure information which is essential for the continued functioning of society. This will inevitably require the ability to work with photons, in particular for transmitting quantum information. Superconducting nanowire single-photon detectors (SNSPDs) are an excellent detector technology for meeting this need. Thus, with this thesis I hope to bring a clearer understanding of the untapped potential and possible pitfalls of these detectors.

Here, I develop a simple electro-thermal model for understanding SNSPDs dynamics, including photon-number resolution (PNR). Then, I experimentally demonstrate PNR in a single-pixel SNSPD, which is a significant advancement for high-speed PNR. Additionally, I experimentally verify the dependence of SNSPD readout pulse rise time on detector length and detector bias current, which is important for understanding how detector design and operation affects the readout signal. Based on anomalously high noise associated with the experimental PNR, I postulate that there could be a below- T_C resistance in these SNSPDs and make predictions for how this would affect detector dynamics. Finally, I measure this resistance experimentally and confirm that there is a nonzero, temperature-dependent below T_C resistance in these SNSPDs.

Based on this work, there are many exciting avenues to explore. The PNR demonstrated

here is an exciting development for the SNSPD community, and an obvious follow up direction is to try to identify a way to improve the PNR. One potential option is to explore the effect of photon time of arrival on detector readout. If two photons arrive simultaneously at the detector, we expect to observe the photon-number dependence described throughout this work. If the two photons arrive far apart in time, it is likely that the detector is no longer sensitive to photons and so will only register a single-photon event. However, it is reasonable to imagine that there is some intermediate region in which the second photon affects the readout pulse, but the effect is not as strong as for two simultaneous photons. This could result in a decrease in overall PNR. A simple experiment to test this would be to repeat the procedure described in Ch. 4 with a short pulse source, such as a <1 ns mode-locked laser.

With the PNR achieved in this thesis, many of the applications that require PNR become more attainable, including linear-optics quantum computing and boson sampling. Proof of principle experiments in these areas should be pursued based on these results. However, there are even simpler experiments that would be interesting to explore, including repeating the experiments described here with a source of known photon-number. For example, measuring the output of a Hong-Ou-Mandel interferometer directly with a single-pixel SNSPD would be an excellent demonstration of this new technology.

Further, the below- T_C resistance measured here is very unexpected and likely to be controversial in the field. Therefore, it is important to verify these results and identify other ways to measure this resistance. One possible method is to examine the falling edge of SNSPD readout pulses, which have a falling time scale of $\tau_R = L_k / R_L$. As described in Ch. 5, for an AC-coupled SNSPD, a below- T_C resistance will manifest mathematically as a larger R_L . By carefully fitting to the falling edge of SNSPD readout pulses, one might extract values for this anomalous resistance.

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Appendix A

Cryocooler System

In this section I describe the cryocooler system used throughout this thesis. I provide a protocol for operating and maintaining the system, as well as a few tips and tricks that allow for easier use in some experiments. Please note that there is a folder on the group drive ([QuantInfoProjects/GroupNotes/Cryostat](#)) that contains more information pertaining to the cryocooler system, including manuals, calibration files, and lists of parts.

A.1 Overview

The cryocooler system is based heavily on a system that was originally developed in Sae Woo Nam's group at NIST (*National Institute of Standards and Technology*) and has a nearly-identical twin in operation at Duke University in Jungsang Kim's group. The system is capable of achieving base temperatures of ~ 850 mK and has the capacity to allow operation of up two SNSPDs operating at 1550 nm and two SNSPDs operating at 700 nm simultaneously. (If light input is not required, the maximum number of detectors operable simultaneously is nine.)

There are three main components to the cryocooler system: the cryostat, the compressor, and the vacuum pump, as shown in Fig. [A.1](#).

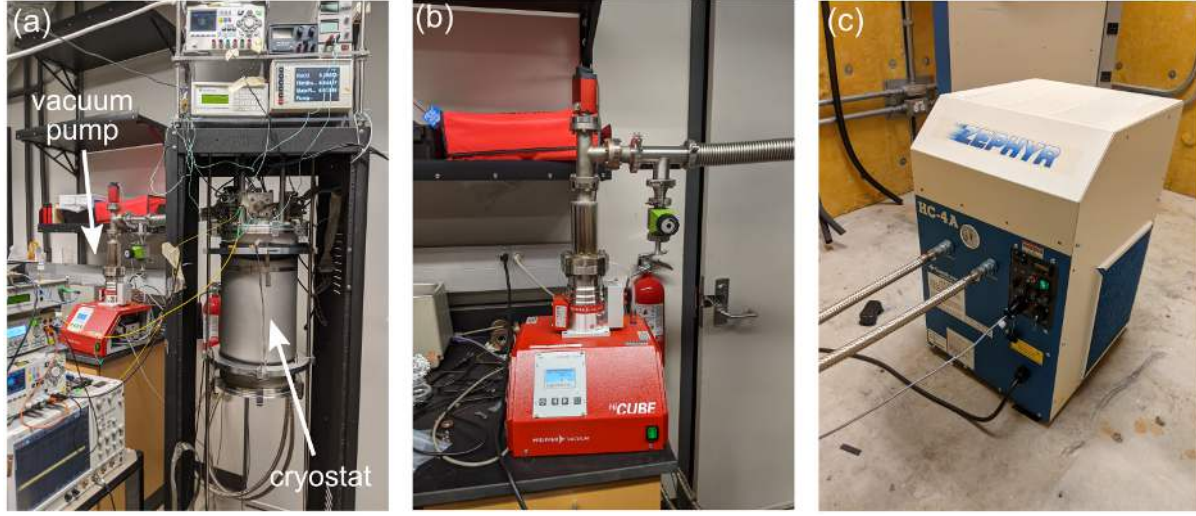


FIGURE A.1: Components of the cryocooler system: (a) cryostat, (b) vacuum pump, (c) compressor.

A.1.1 Cryostat and Compressor

The cryostat is the part of the system that is capable of cooling to cryogenic temperatures and it consists of two separate cooling devices. The first is a Sumitomo RDK-101D cold head which cools from room temperature to ~ 4 K. The cold head has two temperature stages: the upper stage cools to ~ 40 K and the lower stage cools to the base temperature of ~ 4 K. The cold head is powered and provided compressed He by the Sumitomo HC-4A Zephyr compressor. The second cooling device is a Chase Research Cryogenics Single-Stage Sub-Kelvin ^3He Cooler mounted on the lower stage of the cold head. This cryocooler is capable of cooling from ~ 4 K to 850 mK. We mount a copper plate onto the base of the Chase Cryocooler, which constitutes the overall lowest stage of the cryogenic system at 850 mK. This final stage is where all SNSPDs are mounted for operation. Each of the cryogenic stages described here are surrounded by individual radiation shields to minimize the chance of background radiation interfering with the cooling process or the operation of the SNSPDs. The entire cryostat is enclosed in a vacuum-tight cylinder and pumped down to ~ 1 Pa before turning on the compressor to initiate cooling. At base temperature, the cryostat achieves a pressure below 10^{-7} hPa.

The top of the cryostat has 9 SMA and 25 DC vacuum-tight electrical feedthroughs to allow communication with any components inside. Note that as of the writing of this thesis there are 10 SMA feedthroughs, but the 10th is damaged and will need to be replaced to restore functionality. The Chase Cryocooler requires a number of electrical connections to properly function, which practically means there are 6 auxiliary DC lines. The cryostat has been designed to allow for more DC lines to be installed, which is an upgrade that may be done in the future. A pinout diagram for the configuration as of the writing of this thesis is available at QuantInfoProjects/GroupNotes/Cryostat/OSUAssembly/cryostatpinout.pdf.

The cryostat also has fiber optic feedthroughs to allow light to travel to the SNSPDs. There are currently two SMF-28e (1550 nm) fibers labeled "1550-1" and "1550-3" and two SMF-600 (630 nm) fibers labeled "600-1" and "600-2". (Note there were originally four of each, but they have been broken. These broken fibers could be reconnectorized in the future, and so they may still have a connected fiber on either side of the feedthrough.) The fibers are connectorized with FCP-APC fiber on the air side, and PC ferrules on the vacuum side. On the 4 K stage, the fibers are coiled to allow long-wavelength black body radiation to couple out of them.

A.1.2 Vacuum System

For proper operation, the cryostat must be pumped down to a pressure of ~ 1 Pa before commencing the cooling process. This is achieved by enclosing the entire cryostat in a vacuum-compatible cylindrical chamber and pumping said chamber down with a combination Pfeiffer HiCube roughing and turbo pump. There are three main valves within the vacuum system as shown in Fig. A.2. The first is at the top of the cryostat and allows for isolating the pump from the cryostat. The second is a venting pump that connects the turbo directly to the room air. Turbo pumps are easily damaged or destroyed by exposure to sudden increases in pressure, so it is extremely important that this venting valve is never opened unless the turbo pump is off and has stopped spinning completely. The final valve is a secondary venting valve located on the back of the turbo pump itself.

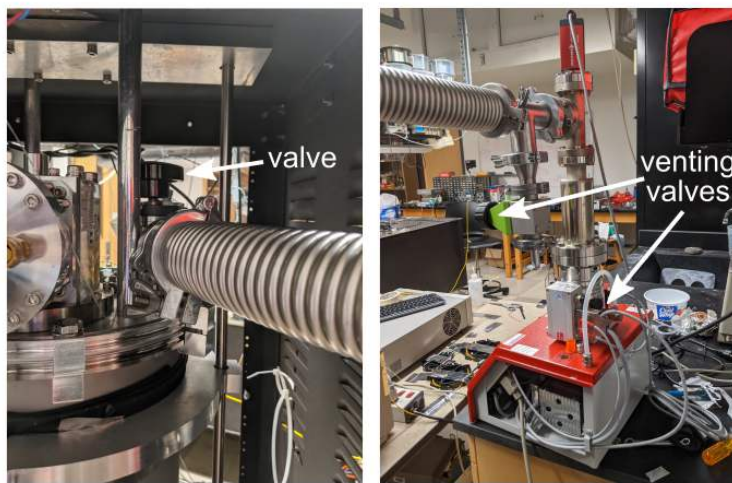


FIGURE A.2: Vacuum valves in the cryostat system. (a) A valve is present at the top of the cryostat that connects it to the rest of the vacuum system. (b) Two venting valves are present.

Unlike many high vacuum systems, thanks to its cryogenic nature this system is relatively robust to contaminants (grease, oils from skin, etc.) that would typically severely hinder vacuum quality. Still, it is important to clean all possible components with a sonicator or methanol before installing them in the cryostat. Additionally, wearing clean gloves whenever working inside the cryostat is important for maintaining the overall health of the system and protecting the worker from any dangerous materials (e.g. cryogenic varnish) used inside the cryostat.

A.1.3 Sensors/Readings Available

Thanks to the electrical feedthroughs mentioned above, there are a number of live readings we can use to monitor the operation of the cryostat. The four main readings are displayed on the Stanford Research Systems CTC100 and are labeled "RuO2", "FilmBurner", "MainPlate", and "Pump". The most useful of these is the RuO2 reading, which provides the temperature at the lowest stage of the cryostat. The others are important for proper cycling of the Chase, as described below. There is also a reading of the Chase switch temperature available on the

Lakeshore 218 Temperature Monitor. For more information, refer to the Chase Cryocooler manual, which is located in the group drive [QuantInfoProjects/GroupNotes/Cryostat/Chase1K/OSU-Operate_CRC4-017.pdf](#).

A.2 Operating Procedure

Typical operation of the cryostat involves first allowing the cold head to cool to its base temperature, and then initiating a cooling cycle with the Chase Cryocooler to reach 850 mK.

Cooling from room temperature to ~ 4 K:

1. Tightly secure all radiation shields within the cryostat.
2. Close large vacuum canister over the cryostat. (This is easiest done with two people, although one person can manage after gaining some experience.)
3. Ensure valve at the top of the cryostat is open and all venting valves are closed.
4. Start the turbo pump by pressing the start button.
5. Once the turbo pump is spinning at 1500 Hz, turn on the compressor.

The system should cool to ~ 4 K within 24 hours. Once it has reached the desired temperature, it is advised to close the valve at the top of the cryostat.

To achieve temperature below 1 K, the Chase Cryocooler must be cycled. This process is easily automated and a program for doing so can be found in [QuantInfoProjects/GroupNotes/InstrumentControlWithPython](#); however, it can also be done manually. More details on this process can be found in the manual.

1. Starting once the Sumitomo cold head has reached its base temperature, heat the Chase pump by applying a voltage of 25V until the pump temperature reaches 45 K. (Note that as of the writing of this thesis, the pump temperature probe is not functional. However, in my experience it is adequate to heat the pump for 12 minutes to achieve this temperature.)

2. Stop heating the pump and wait for the coldfinger (RuO₂) temperature to get below the liquid-He transition temperature (4.2 K). The lower this temperature is allowed to fall, the longer the Chase Cryocooler will hold its base temperature. In my experience, waiting for an hour allows for a hold time of ~ 8 hours.
3. Turn on the switch by applying a voltage of 4 V. The coldfinger will rapidly begin cooling. Absolute base temperature is typically achieved in about 20 minutes.
4. The Chase Cryocooler should hold its base temperature for about 8 hours. After this the cycle can be repeated.

Warming up is the easiest process to achieve and can be done regardless of current temperature of the system.

1. Turn off the compressor. (For fastest warm up, make sure the valve is closed at the top of the cryostat.)
2. Once the coldest stage of the cryostat (RuO₂ reading) has reached room temperature, turn off the turbo pump. It typically takes more than 12 hours for the whole system to warm up.
3. Once the turbo pump has stopped spinning, open the valve at the top of the cryostat.
4. Open either of the venting valves to bring the system to atmospheric pressure.

A.3 Tips and Tricks

In my years working with the cryostat, there are a few tricks I have learned that can prove useful.

Making Good Thermal Contact

For the system to cool properly, it is important to make sure to make good thermal connections in the appropriate places. For example, when mounting a detector on the 1 K stage, it is important it makes very good thermal contact with the stage to cool properly. I typically use a very thin layer of vacuum grease (Apiezon N) between any components and their respective stages to ensure a good thermal connection. Additionally, it is best to use spring/lock washers with all screws.

On the other hand, it is important not to accidentally allow unintended components to interact thermally. For example, if a cable or similar component is oriented so that it is touching one of the radiation shields, it is very likely to sabotage the cooling of the entire system. When closing up the system, check carefully for stray thermal connections when attaching each of the radiation shields.

When possible, it is useful to use superconducting coaxial cables to route between the 4 K and 1 K plates. These minimize heat transfer down to the 1 K plate. We currently have some designed to go from SMA (4 K plate) to SMP (the NIST detectors) which were constructed using an in-house electroplating process (DeBry and Lafyatis, 2018).

A.3.1 Faster Warm Up

Although this is counter-intuitive, I have experienced that when trying to warm up, making sure the valve at the top of the cryostat is closed drastically improves warmup times. With the valve closed, the system typically takes less than 24 hours to warm from base temperature to room temperature. When I have forgotten and left the valve open, the cryostat is typically still at liquid N temperatures after 24 hours.

A.3.2 Slower Warming from 850 mK to 4 K

For many applications, it is useful to be able to slowly sweep through temperature. However, the cryocooler system does not allow for holding a specific temperature that is not the base temperature of the Sumitomo cold head or the Chase Cryogenics Cryocooler. Further, specifically when warming from 850 mK to 4 K, the design of the Chase Cryocooler causes it to warm very rapidly, typically on the timescale of a few minutes. I have found that severing the thermal link between the pump and the 1 K stage by turning off the switch results in a significantly slower warm up time. This technique is used widely throughout this thesis any time measurements are taken as a function of temperature.

A.4 General Maintenance

The Sumitomo cold head requires maintenance at the factory every 10000 hours. When it is getting close to maintenance time, it sometimes begins to make a grinding noise while pumping He. Examples of this noise can be found on the group drive. Additionally, the compressor has a hour counter to help keep track of runtime.